

REINFORCE PRO MAX: CAUSAL PREFIX FILTERING AND ADAPTIVE ADVANTAGE NORMALIZATION FOR CRITIC-FREE LLM REINFORCEMENT LEARNING

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ABSTRACT

Critic-free reinforcement learning for language models must control both token-mass imbalance in sampled advantages and the mismatch between rollout and training policies. We present REINFORCE Pro Max, which combines leave-one-out advantages, sign-preserving asymmetric normalization, and a prefix-cumulative importance-sampling gate. The gate is computed from the cached rollout/snapshot ratio but gradients flow only through the current/snapshot PPO ratio. On a finite autoregressive tree, we prove an exact decomposition of the implemented standard-PPO objective into the raw surrogate, gate and normalization corrections, and a nonnegative clipping deduction. A finite-tree adaptive bound then gives a conditional sufficient certificate for reward improvement. The mathematical statements and their finite-tree proofs are checked in Lean; the formalization does not claim to verify floating-point neural-network training. The current artifact contains structural synthetic checks only and does not report an LLM benchmark.

1 PROBLEM AND SCOPE

Critic-free objectives such as RLOO and GRPO are widely used for reasoning tasks (Ahmadian et al., 2024; Shao et al., 2024), while PPO remains a standard clipped policy-gradient baseline (Schulman et al., 2017). Recent long-horizon trust-region analyses motivate retaining per-position causal structure (Li et al., 2025). LLM-RL rollouts are generated by π_{roll} while the actor is updated from a frozen snapshot π_{old} . For response $y = (y_1, \dots, y_T)$, write $\rho_t = \pi_{\theta}(y_t | c_t) / \pi_{\text{roll}}(y_t | c_t)$ and $w_t = \pi_{\text{old}}(y_t | c_t) / \pi_{\text{roll}}(y_t | c_t)$. The analyzed configuration is standard PPO (`policy_loss_type=ppo`), `vllm_is_correction_type=reinforce_pro`, no dual clip, no KL or entropy term, and sparse terminal reward with zero KL coefficient. These restrictions are essential: optional dual clipping and auxiliary losses change the objective.

We study a finite response alphabet and finite horizon with mutual support, bounded reward $|R| \leq R_{\text{max}}$, and explicitly stated trust-region or moment assumptions. The results are conditional certificates, not guarantees that an arbitrary trained neural policy satisfies those assumptions.

2 METHOD

For a prompt group of n responses, the raw leave-one-out signal is $A_i = R_i - (n-1)^{-1} \sum_{j \neq i} R_j$. After token expansion, positive and negative active tokens need not have equal mass. Let P, N be the positive and negative token sets. The ideal ProMax factors are chosen so that $\sum_P \alpha A + \sum_N \beta A = 0$ and $|P|^{-1}|N|^{-1}$ -free second-moment normalization equals one on nonzero tokens, with $\alpha, \beta > 0$. Thus signs and within-sign ordering are preserved. The implementation adds an epsilon, cap, clamp, and fallback; the formal result treats their moment effects as explicit residuals.

For each active token, the prefix proxy is

$$M_t^{\text{pre}} = \mathbf{1} \left\{ \ell_- \leq \frac{1}{t} \sum_{s \leq t} \log w_s \leq \ell_+ \right\}, \quad w_t = \exp(\log \pi_{\text{old}} - \log \pi_{\text{roll}}).$$

The implemented loss is

$$\mathcal{L} = \text{MaskedMean}_t \left[M_t^{\text{pre}} \text{stopgrad}(w_t) \left(-\min(r_t \hat{A}_t, \text{clip}(r_t) \hat{A}_t) \right) \right], \quad r_t = \frac{\pi_\theta}{\pi_{\text{old}}}.$$

The prefix gate, cached importance coefficient, action mask, and normalized advantage are detached. Only r_t carries policy gradient.

3 FORMAL RESULTS

Let k index represented response/token positions, let $\omega_k \geq 0$ be the chosen reduction weight, M_k the gate, r_k and w_k the two probability ratios, A_k the raw signal, and H_k the implemented normalized signal. For standard PPO define

$$U = \sum_k \omega_k r_k w_k A_k, \quad G = \sum_k \omega_k r_k w_k (1 - M_k) A_k, \quad N = \sum_k \omega_k r_k w_k M_k (H_k - A_k),$$

and

$$C = \sum_k \omega_k M_k w_k (r_k H_k - \min(r_k H_k, \text{clip}(r_k) H_k)).$$

Then the finite-batch identity is

$$-\mathcal{L} = U - G + N - C, \quad C \geq 0.$$

The gate and normalization terms are signed corrections. The nonnegativity of C is a standard-PPO fact and does not extend to dual clip; a Lean-checked counterexample is included in `formal/ProMaxObjective.lean`.

For a finite autoregressive tree, if every rollout-reachable history of length less than T satisfies either a local TV bound ε or a local reverse-KL bound δ , the future continuation discrepancy after history length t is bounded by

$$\min\{1, (T - t)\varepsilon, \sqrt{(T - t)\delta/2}\}.$$

With bounded rewards, summing these continuation errors bounds the difference between the raw surrogate and the true reward change. Consequently, a corrected empirical ProMax gain exceeding the explicit gate, normalization, clipping, sampling, and adaptive remainder terms is sufficient for positive true reward gain. The statement is for independent evaluation blocks and a fixed finite candidate family, or a pre-specified class with a fixed finite cover.

The finite-tree identities, adaptive recursion, RLOO moments, reduction bounds, and binary-length covariance conditions are checked by the Lean files listed in `formal/PROOF_AUDIT.md`. The audit currently covers 31 labeled mathematical environments and 911 project theorems.

For binary rewards, the ideal normalization has two class amplitudes whose ratio is the ratio of class-conditional mean response lengths. With the actual epsilon stabilizer, cap, clamp, fallback, and token denominator, the complete on-policy update remains a positive multiple of the success-probability gradient when active length depends only on the reward class and both classes have positive mass. If length also varies within a reward class, the update decomposes into this class multiplier plus a within-class covariance error; a strict local-improvement conclusion then requires an explicit covariance margin. The theorem does not claim that arbitrary content-dependent lengths or an arbitrary Adam step improve reward.

4 IMPLEMENTATION CORRESPONDENCE

The code path uses active-token masking and distinguishes the differentiable PPO ratio from the detached rollout correction. Standard PPO is fixed to the configuration above; GSPO, dual clip, KL loss, entropy loss, MoE auxiliary loss, reward shaping, and dynamic group filtering require separate objective terms and are outside the stated certificate. Variable-length dynamic microbatch response-count weighting does not automatically equal a global token mean, so the population reduction must be reported explicitly.

5 REINFORCE MAX

For response i , define $A_i = R_i - b_i$. Sparse terminal reward with $\gamma = 1$ expands to the same signal on every active token. Let $m_{i,t}$ be its action mask and let $P = \{(i, t) : A_i > 0, m_{i,t} = 1\}$ and $N = \{(i, t) : A_i < 0, m_{i,t} = 1\}$. The ideal asymmetric factors $\alpha, \beta > 0$ are chosen from the token masses and second moments so that

$$\sum_P \alpha A_i + \sum_N \beta A_i = 0, \quad \frac{1}{|P| + |N|} \left(\sum_P (\alpha A_i)^2 + \sum_N (\beta A_i)^2 \right) = 1.$$

The normalized signal is $H_{i,t} = \alpha A_i$ on P , βA_i on N , and zero on raw zero tokens. Consequently signs and within-sign ordering are preserved while positive and negative token mass are balanced. The code adds an epsilon, cap, clamp, and fallback; their effects are retained as explicit normalization residuals in the objective decomposition.

Proposition 5.1 (Binary class scaling). *For binary rewards with positive and negative class masses and class-conditional mean active lengths $\bar{L}_+$ and $\bar{L}_-$, the ideal scale ratio satisfies $\alpha/\beta = \bar{L}_-/\bar{L}_+$. Thus the token-weighted update is a positive class-dependent multiple of the success-probability direction when length is constant within each reward class.*

When response length varies within a class, the update contains a covariance term. A strict local-improvement statement therefore requires a covariance margin; the theorem does not assert that arbitrary content-dependent lengths or an arbitrary Adam step improve reward. The Lean files `BinaryLength.lean`, `BinaryUpdate.lean`, and `RLOOCovariance.lean` verify these distinctions.

6 REINFORCE PRO: CAUSAL PREFIX GATE

The cached correction uses the old-to-rollout ratio w_t . For thresholds $0 < \ell_- < 1 < \ell_+$, define the detached prefix gate

$$M_t = \mathbf{1} \left\{ \ell_- \leq \exp \left(\frac{1}{\nu_t} \sum_{s \leq t} m_s \log w_s \right) \leq \ell_+ \right\}, \quad \nu_t = \max \left(1, \sum_{s \leq t} m_s \right).$$

The implemented standard-PPO loss is

$$\mathcal{L} = - \sum_{i,t} \omega_{i,t} M_{i,t} \text{stopgrad}(w_{i,t}) \min\{r_{i,t} H_{i,t}, \text{clip}(r_{i,t}) H_{i,t}\},$$

where ω includes the detached action mask and selected active-token reduction. The gate, w , H , and mask are detached; only the current PPO ratio r carries the actor gradient.

Lemma 6.1 (Prefix stability). *If each represented prefix ratio lies in $[\ell_-, \ell_+]$ and one additional log-ratio perturbation is bounded by b , the next prefix geometric mean changes by at most $b/(t+1)$ in log scale. Therefore a gate crossing can be localized to a suffix after the first rejected prefix.*

Theorem 6.2 (Causal continuation bound). *Under the local trust-region assumption, the total variation between the continuation distributions after a history of depth t is at most*

$$d_t \leq \min \left\{ 1, (T-t)\varepsilon, \sqrt{(T-t)\delta/2} \right\}.$$

For bounded terminal rewards, the corresponding continuation-value error is at most $2R_{\max} d_t$. Summing over reachable prefixes yields the adaptive remainder used in the performance certificate.

The causal gate differs from token-level TIS and sequence-level GSPO. A token outside the range is filtered according to its own prefix history, so early drift propagates forward while a late violation affects only a suffix. The implementation uses detached coefficients, avoiding an unintended gradient through the rollout correction or through the gate.

7 UNIFIED OBJECTIVE IDENTITY

Let k enumerate represented active positions, with nonnegative reduction weights ω_k , gate M_k , raw signal A_k , normalized signal H_k , and ratios r_k, w_k . Define

$$\begin{aligned} U &= \sum_k \omega_k r_k w_k A_k, & G &= \sum_k \omega_k r_k w_k (1 - M_k) A_k, \\ N &= \sum_k \omega_k r_k w_k M_k (H_k - A_k), & C &= \sum_k \omega_k M_k w_k (r_k H_k - \min(r_k H_k, \text{clip}(r_k) H_k)). \end{aligned}$$

Proposition 7.1 (Finite-batch decomposition). *For standard PPO,*

$$-\mathcal{L} = U - G + N - C, \quad C \geq 0,$$

and therefore $|-\mathcal{L} - U| \leq |G| + |N| + C$. The identity holds for every realized batch, including random lengths and cached policy mismatch.

The three correction terms have different meanings: G is the signed cost of prefix rejection, N is the signed cost of numerical normalization, and C is the standard-PPO clipping deduction. The latter nonnegativity is not true for dual clipping; the checked counterexample uses advantage -1 , current ratio 4, clipped ratio $6/5$, and dual coefficient 3, yielding deduction -1 .

Theorem 7.2 (Finite-sample reward certificate). *For a finite candidate family fixed before evaluation, K independent RLOO blocks, and minimum active length ℓ , let $\hat{\kappa}_j$ be the self-normalized corrected increment for candidate q_j , and let B_j be the summed continuation remainder from the causal bound. With probability at least $1 - \sum_j v_j / [K(n\ell)^2 \eta_j^2]$, simultaneously for all candidates,*

$$J(q_j) - J(p) \geq \mu_L (\Delta S_{\text{PM}} + \Delta G - \Delta N + \Delta C - \eta_j) - B_j.$$

Thus a positive corrected gain exceeding sampling, gate, normalization, clipping, and adaptive remainders is a sufficient certificate for true reward improvement. It is conditional on the stated finite-tree and trust-region assumptions, not a guarantee for an arbitrary neural policy.

8 SAMPLING, REDUCTION, AND IMPLEMENTATION CORRESPONDENCE

RLOO is unbiased for the response-level policy-gradient direction under conditional i.i.d. responses. For a response contribution Z and active length L , response-mean reduction targets $\mathbb{E}[Z/L]$, whereas global token reduction targets $\mathbb{E}[Z]/\mathbb{E}[L]$. Their difference is

$$\mathbb{E}[Z/L] - \frac{\mathbb{E}[Z]}{\mathbb{E}[L]} = -\frac{\mathbb{E}[(L - \mathbb{E}[L])Z/L]}{\mathbb{E}[L]}.$$

This length covariance is why the theorem states the selected reduction explicitly. The standard PPO code path uses active-token mean; dynamic response-count replay weighting does not automatically reconstruct a global token mean for variable lengths. GSPO, dual clip, KL loss, entropy loss, reward shaping, and auxiliary losses are separate objectives outside this certificate.

9 PROOF ROADMAP AND FORMAL SCOPE

The main proof chain is: RLOO cancellation and moments; token expansion and normalization; prefix-gate stability and continuation discrepancy; exact finite-batch decomposition; self-normalized sampling concentration; and the finite-tree adaptive reward bound. Each theorem is mapped to a Lean declaration in `formal/PROOF_AUDIT.md`. The formal development checks exact finite models and real arithmetic, including the dual-clip counterexample and the binary-length covariance boundary. It does not claim to formalize floating point, GPU kernels, Adam, Ray/vLLM scheduling, or general neural-network optimization.

10 PRELIMINARIES

10.1 AUTOREGRESSIVE GENERATION AND OBJECTIVE

For the finite-horizon identities below, T denotes a fixed deterministic horizon. Variable-length responses are represented by padding with an absorbing end-of-sequence token and by zeroing padded

action terms with the corresponding action mask. The post-termination transition is deterministic and shared by both policies, so its likelihood ratio is 1; this is the convention used whenever a common horizon appears in a sum or product.

A (stochastic) policy π generates a response $y = (y_1, \dots, y_T)$ conditioned on a prompt x . The induced trajectory probability admits the autoregressive factorization

$$P^\pi(y \mid x) = \prod_{t=1}^T \pi(y_t \mid x, y_{<t}).$$

For each position t , we define the context

$$c_t := (x, y_{<t}) = (x, y_{<t}).$$

The induced *context visitation distribution* (the marginal distribution of length- $(t-1)$ prefixes, i.e. contexts before token t) is

$$d_t^\pi(c_t) = P(x) P^\pi(y_{<t} \mid x) = P(x) \prod_{s=1}^{t-1} \pi(y_s \mid c_s).$$

Given a scalar reward $R(x, y)$, the objective of the current policy π_θ is

$$J(\pi_\theta) = \mathbb{E}_{x \sim P(x), y \sim \pi_\theta(\cdot \mid x)} [R(x, y)].$$

10.2 NOTATION

Policies. We consider:

- π_{roll} : the *behavior* (rollout) policy used to generate samples; it is fixed during each policy update;
- π_θ : the *training* policy being optimized by gradient steps.
- π_{old} : the frozen PPO snapshot used by the clipped current-policy ratio; it is an implementation object and need not equal π_{roll} .

This notation follows long-horizon trust-region analyses for LLM-RL (e.g., Li et al. (2025)).

Importance ratios. The per-token importance ratio and its log-ratio are

$$\rho_t := \frac{\pi_\theta(y_t \mid c_t)}{\pi_{\text{roll}}(y_t \mid c_t)}, \quad (1)$$

$$\log \rho_t = \log \pi_\theta(y_t \mid c_t) - \log \pi_{\text{roll}}(y_t \mid c_t). \quad (2)$$

Rollout-time anchoring. In practice, inference and training stacks can differ. The theoretical ρ_t uses rollout-time cached log-probabilities $\log \pi_{\text{roll}}(y_t \mid c_t)$. The implementation obtains the corresponding correction through the factorization $\rho_t^{\theta/\pi_{\text{old}}} = r_t^{\theta/\pi_{\text{old}}} w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$; see Appendix B.1 for practical mismatch modes and implementation mappings.

10.3 DIVERGENCE MEASURES

We use token-level divergence measures between conditional token distributions, following the trust-region literature for long-horizon LLM-RL (Li et al., 2025). For two policies π and π' , define the token-level total variation and KL divergences at context c_t :

Definition 10.1 (Token-Level Divergences). For a context $c_t = (x, y_{<t})$ and two policies π and π' ,

$$D_{\text{TV}}^{\text{tok}}(c_t; \pi, \pi') := D_{\text{TV}}(\pi(\cdot \mid c_t) \parallel \pi'(\cdot \mid c_t)) = \frac{1}{2} \sum_v |\pi(v \mid c_t) - \pi'(v \mid c_t)|,$$

$$D_{\text{KL}}^{\text{tok}}(c_t; \pi, \pi') := D_{\text{KL}}(\pi(\cdot \mid c_t) \parallel \pi'(\cdot \mid c_t)) = \sum_v \pi(v \mid c_t) \log \frac{\pi(v \mid c_t)}{\pi'(v \mid c_t)}.$$

When the policy pair is clear from context, we write $D_{\text{TV}}^{\text{tok}}(c_t)$ and $D_{\text{KL}}^{\text{tok}}(c_t)$.

Definition 10.2 (Max Token-Level Divergences and Expected Per-Position TV). For the policy pair $(\pi_{\text{roll}}, \pi_\theta)$, define:

$$\begin{aligned}\epsilon &:= D_{\text{TV}}^{\text{tok}, \max} := \max_{t, c_t} D_{\text{TV}}^{\text{tok}}(c_t), \\ \delta &:= D_{\text{KL}}^{\text{tok}, \max} := \max_{t, c_t} D_{\text{KL}}^{\text{tok}}(c_t), \\ \bar{D}_t &:= \mathbb{E}_{c_t \sim d_t^{\pi_{\text{roll}}}} [D_{\text{TV}}^{\text{tok}}(c_t)].\end{aligned}$$

10.4 ASSUMPTIONS

The following assumptions are used throughout the paper:

Assumption 10.3 (Mutual Policy Support). For all contexts c_t and tokens y_t in the vocabulary,

$$\pi_{\text{roll}}(y_t | c_t) > 0 \iff \pi_\theta(y_t | c_t) > 0.$$

That is, the two conditional policies have mutual support. This makes the per-token importance ratio ρ_t and its logarithm in Eqs. equation 1–equation 2 well-defined on every sampled token, and permits the change-of-measure identities used below (in particular $\mathbb{E}_{\pi_{\text{roll}}}[\rho_t | c_t] = 1$ and the TV/KL ratio formulas). All ratio identities are interpreted on the rollout support $\{y_t : \pi_{\text{roll}}(y_t | c_t) > 0\}$. If both policies assign zero mass to a token, the ratio at that token is assigned an arbitrary harmless value (the guarded finite KL formalization uses 1), because such tokens contribute zero to rollout expectations and to the KL/TV sums. Proofs that work directly with token ratios restrict them to the positive rollout support. Thus the assumption does not require strict positivity on the whole vocabulary.

Assumption 10.4 (Bounded Rewards). There exists $R_{\max} < \infty$ such that $|R(x, y)| \leq R_{\max}$ for all prompt-response pairs (x, y) .

Assumption 10.5 (Finite Sequence Length). There is a fixed finite generation cap T . Every response has at most T active actions and is extended to this common horizon by the shared absorbing EOS convention above.

Remark 10.6 (Three-policy ratio decomposition in the implementation). The mathematical statements below use the behavior/training pair $(\pi_{\text{roll}}, \pi_\theta)$. The PPO implementation also has a frozen snapshot π_{old} and therefore uses two ratios:

$$r_t^{\theta/\pi_{\text{old}}} := \frac{\pi_\theta(y_t | c_t)}{\pi_{\text{old}}(y_t | c_t)}, \quad w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} := \frac{\pi_{\text{old}}(y_t | c_t)}{\pi_{\text{roll}}(y_t | c_t)}.$$

On their common support,

$$\rho_t^{\theta/\pi_{\text{roll}}} := \frac{\pi_\theta(y_t | c_t)}{\pi_{\text{roll}}(y_t | c_t)} = r_t^{\theta/\pi_{\text{old}}} w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}.$$

The corresponding log-probability implementation identity is checked by `ImplementationBridge.current_rollout_ratio_factorization`; it is the exponentiated form of the three-policy decomposition used below. The prefix gate in the current implementation is computed from $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$, using cached rollout log-probabilities and the old-policy log-probabilities; the PPO surrogate is computed from $r_t^{\theta/\pi_{\text{old}}}$. When $\pi_{\text{old}} = \pi_{\text{roll}}$ at the beginning of an update, $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} = 1$ and the PPO ratio equals the theoretical behavior-to-current ratio. For stale snapshots, the prefix results below apply directly to the pair $(\pi_{\text{roll}}, \pi_{\text{old}})$; transferring them to the current pair additionally requires a bound on the PPO update.

10.5 THE SURROGATE OBJECTIVE AND ERROR DECOMPOSITION

Samples are generated from a behavior policy π_{roll} that generally differs from π_θ . Following Kakade & Langford (2002), the standard surrogate objective uses the per-step advantage under π_{roll} ,

$$A_t^{\pi_{\text{roll}}}(c_t, y_t) := \mathbb{E}_{\pi_{\text{roll}}}[R(x, y) | c_t, y_t] - \mathbb{E}_{\pi_{\text{roll}}}[R(x, y) | c_t].$$

This yields the per-step surrogate

$$\mathcal{L}_{\pi_{\text{roll}}}(\pi_\theta) = \sum_{t=1}^T \mathbb{E}_{c_t \sim d_t^{\pi_{\text{roll}}}} \left[m_t(c_t) \mathbb{E}_{y_t \sim \pi_{\text{roll}}(\cdot | c_t)} [\rho_t A_t^{\pi_{\text{roll}}}(c_t, y_t)] \right], \quad (3)$$

where $m_t(c_t) \in \{0, 1\}$ is the predictable action mask from Definition 10.7. In the unpadded fixed-horizon model $m_t \equiv 1$; with EOS padding it removes post-termination contexts, whose shared absorbing transition has zero advantage and unit ratio. The ratio ρ_t is defined in Eq. equation 1.

Definition 10.7 (Trajectory-Level Surrogate). Let b be a baseline that is independent of the sampled trajectory y conditional on the prompt x (e.g., a prompt-level baseline computed from other independent rollout samples in the same prompt group). The rollout policy and the baseline sampling law are fixed during the update. Let $m_t = m_t(c_t) \in \{0, 1\}$ denote the predictable action mask: it equals one while generation is active, including the action that emits EOS, and zero at subsequent padding positions. Define the trajectory-level advantage

$$A := R(x, y) - b.$$

The unclipped, unnormalized trajectory-level surrogate underlying critic-free implementations is

$$\tilde{\mathcal{L}}_{\pi_{\text{roll}}}(\pi_{\theta}) := \mathbb{E}_{\pi_{\text{roll}}} \left[A \cdot \sum_{t=1}^T m_t \rho_t \right].$$

Proposition 10.8 (Exact Centered Surrogate Equivalence). *For finite prompt, action, and continuation supports, under mutual policy support and the finite/EOS-padded horizon convention, assume that the expectations below exist and that b and m_t satisfy Definition 10.7. Then, for any current policy π_{θ} ,*

$$\tilde{\mathcal{L}}_{\pi_{\text{roll}}}(\pi_{\theta}) - \tilde{\mathcal{L}}_{\pi_{\text{roll}}}(\pi_{\text{roll}}) = \mathcal{L}_{\pi_{\text{roll}}}(\pi_{\theta}) = \mathbb{E}_{\pi_{\text{roll}}} \left[R(x, y) \sum_{t=1}^T m_t (\rho_t - 1) \right]. \quad (4)$$

The subtracted quantity is constant during the policy update; the equality does not require $\pi_{\theta} = \pi_{\text{roll}}$.

The proof is in Appendix E.1. This is a population identity; individual Monte Carlo estimates need not coincide. The random-length batch reductions used by ProMax are retained in ??.

Remark 10.9 (Gradient Interpretation and ProMax Modifications). Where differentiable, Eq. equation 4 implies equality of the gradients of the two surrogates while keeping the rollout distribution fixed. At the on-policy evaluation point, the REINFORCE identity gives

$$\mathbb{E}_{\pi_{\theta}} \left[A \cdot \sum_{t=1}^T m_t \nabla_{\theta} \log \pi_{\theta}(y_t | c_t) \right] = \nabla_{\theta} J(\pi_{\theta}),$$

under the usual score-function differentiation conditions. Away from that point, equality of the two surrogate gradients does not equate either gradient with $\nabla_{\theta} J(\pi_{\theta})$. The actual ProMax loss additionally applies adaptive normalization, an action-dependent prefix gate, PPO clipping, and the selected batch reduction. These operations require their own analysis; the exact finite-batch decomposition in ?? retains all of them. In particular, detaching a prefix gate does not make it independent of the sampled action.

The approximation error of the per-step surrogate is

$$\text{Error}(\pi_{\theta}) := J(\pi_{\theta}) - J(\pi_{\text{roll}}) - \mathcal{L}_{\pi_{\text{roll}}}(\pi_{\theta}).$$

Under the finite-horizon Performance Difference Identity (Kakade & Langford, 2002), and assuming finite context/token supports so that the conditional expectations are finite sums, this error decomposes as:

$$|\text{Error}(\pi_{\theta})| \leq \sum_{t=1}^T 2 \|g_t\|_{\infty} \cdot \|d_t^{\pi_{\theta}} - d_t^{\pi_{\text{roll}}}\|_{\text{TV}}, \quad (5)$$

where

$$g_t(c_t) := m_t(c_t) \mathbb{E}_{y_t \sim \pi_{\theta}(\cdot | c_t)} [A_t^{\pi_{\text{roll}}}(c_t, y_t)]$$

is the expected advantage shift. Classical bounds on Eq. equation 5 scale as $O(T^2)$ (Kakade & Langford, 2002; Achiam et al., 2017), rendering them theoretically vacuous for long-horizon LLM tasks with responses of thousands of tokens. Recent work (Li et al., 2025) derives tighter bounds,

including the Pinsker-Marginal bound ($O(T^{3/2})$), the Mixed bound ($O(T)$), and the data-dependent Adaptive bound.

The finite-sum error derivation and its formalization are recorded in Appendix E.1.

The Adaptive bound of Li et al. (2025) is particularly relevant to our work. We give a complete finite-tree proof, including the linear future-TV route, to fix its constants and connect it to the same surrogate used by the method.

Theorem 10.10 (Finite-tree Adaptive remainder bound). *For a finite prompt distribution and finite vocabulary, let both policies be normalized history-dependent kernels satisfying Assumption 10.3 and the fixed/EOS-padded horizon convention. Suppose $|R| \leq R_{\max}$ with $R_{\max} \geq 0$, and let ϵ, δ bound the local TV and reverse KL at every context before horizon T , as in Definition 10.2. Then*

$$B_{\text{Adap}}^* = 4R_{\max} \sum_{t=1}^T \bar{D}_t \cdot \min\left(1, (T-t)\epsilon, \sqrt{\frac{(T-t)\delta}{2}}\right), \quad (6)$$

satisfies

$$|J(\pi_\theta) - J(\pi_{\text{roll}}) - \mathcal{L}_{\pi_{\text{roll}}}(\pi_\theta)| \leq B_{\text{Adap}}^*, \quad J(\pi_\theta) - J(\pi_{\text{roll}}) \geq \mathcal{L}_{\pi_{\text{roll}}}(\pi_\theta) - B_{\text{Adap}}^*.$$

The same bound controls the absolute reward-weighted trajectory remainder in Definition E.7.

Proof. The exact history-tree error recursion separates the current token’s probability shift from the difference in future expected rewards. The latter is bounded by $2R_{\max}$ times the future trajectory TV, which is at most each of $1, (T-t)\epsilon$, and $\sqrt{(T-t)\delta/2}$. Summing the local interaction under rollout prefixes gives Eq. equation 6. The full recursion and finite prompt averaging are proved in Appendix D.2 and checked in `formal/AdaptiveBound.lean`. $\square$

The factor $R_{\max}$ is required for general bounded rewards; $R_{\max} = 1$ recovers the normalized-reward form. This result retains the per-position expected TV and the remaining future horizon. It motivates causal prefix filtering, while the sampled gate’s own guarantees are stated separately in ??; a sampled acceptance event does not establish the population TV/KL assumptions of Theorem 10.10.

10.6 EXISTING IMPORTANCE SAMPLING CORRECTION METHODS

Several methods have been proposed to stabilize policy-gradient updates by constraining the per-token importance ratio ρ_t between the training policy π_θ and the behavior policy π_{roll} . We use the per-token log-ratio $\log \rho_t$ defined in Eq. equation 2.

Table 1 summarizes these stabilization mechanisms under a unified notation.

Table 1: Existing trust-region and IS stabilization methods in critic-free LLM-RL.

Method	IS Statistic	Truncation	Granularity
TIS	ρ_t	$w_t^{\text{tok}} = \text{clip}(\rho_t, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}})$	Token
IcePop	ρ_t	$M_t^{\text{tok}} = \mathbb{I}[\rho_t \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]]$	Token
TRM	ρ_t	$M^{\text{TRM}}(x, y) = \mathbb{I}[\max_t D_{\text{KL}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_\theta) \leq \delta_{\text{TRM}}]$	Sequence
GSPO	ρ^{seq}	$w^{\text{seq}} = \text{clip}(\rho^{\text{seq}}, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}})$	Sequence

The table uses the theoretical behavior-to-training ratio for a common notation. In the PPO implementation, the corresponding mismatch correction weights are evaluated from $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$, while the differentiable PPO ratio is $r_t^{\theta/\pi_{\text{old}}}$, as specified above.

Here ρ^{seq} is the length-normalized sequence likelihood ratio (a geometric mean of token ratios),

$$\begin{aligned} \rho^{\text{seq}} &:= \left(\frac{P^{\pi_\theta}(y \mid x)}{P^{\pi_{\text{roll}}}(y \mid x)} \right)^{\frac{1}{T}} \\ &= \left(\prod_{t=1}^T \rho_t \right)^{\frac{1}{T}} = \exp\left(\frac{1}{T} \sum_{t=1}^T \log \rho_t\right), \end{aligned}$$

where T is the fixed (or EOS-padded) horizon used for the comparison. For an unpadded response, this reduces to its actual length $T = |y|$. This definition matches the sequence-level ratio used in GSPO (Zheng et al., 2025) (denoted $s_i(\theta)$ in the GSPO paper) up to notation.

TRM (Trust Region Masking, Li et al. (2025)) uses a binary *sequence-level* mask $M^{\text{TRM}}(x, y)$ to discard trajectories that violate the trust region. A standard (max-based) criterion is

$$M^{\text{TRM}}(x, y) := \mathbb{I}\left[\max_t D_{\text{KL}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_\theta) \leq \delta_{\text{TRM}}\right].$$

A comparative discussion of these mechanisms and their limitations is provided in ??.

Remark 10.11 (Context Shift Propagation). Fix a prompt and a finite vocabulary, with normalized policies on complete prefix contexts. Let s be the first position of disagreement: the policies agree at every rollout-reachable context before s , and $D_{\text{TV}}^{\text{tok}}(c_s; \pi, \pi') > 0$ at some context with $d_s^\pi(c_s) > 0$. Then

$$\|d_{s+1}^\pi - d_{s+1}^{\pi'}\|_{\text{TV}} = \mathbb{E}_{c_s \sim d_s^\pi} [D_{\text{TV}}^{\text{tok}}(c_s; \pi, \pi')] > 0.$$

Moreover, full-prefix TV is nondecreasing with prefix length, so every later context distribution retains at least this discrepancy. Marginalizing the last token recovers the preceding complete prefix; the triangle inequality therefore prevents its TV from increasing under that marginalization. No support restriction on subsequent policy kernels is needed, including after a shared absorbing EOS transition. The equality, monotonicity, and first-disagreement implication are checked by `prefixTV_next_of_equal_prefix_mass`, `prefixTV_mono`, and `first_disagreement_propagates` in `formal/TreeTV.lean`. This statement concerns full histories; a representation that discards earlier tokens need not retain their discrepancy. See also Li et al. (2025).

11 THEORY SCOPE AND LIMITATIONS

This is a theory paper. Deterministic structural checks for normalization and prefix masking are included as consistency checks, rather than as an empirical benchmark. The paper makes no claim about benchmark accuracy, sample efficiency, or wall-clock performance. The formal proofs use exact real arithmetic on finite models; they do not formalize floating-point overflow, GPU kernels, Adam, Ray/vLLM execution, or the assumptions of a general neural policy. These boundaries are part of the theorem statements and determine the conditions under which the results apply.

AI USE STATEMENT

Generative AI tools assisted with code navigation, proof-audit organization, and language editing. All mathematical statements, source changes, and reported checks were reviewed against the repository. The authors take responsibility for the final content.

REPRODUCIBILITY STATEMENT

The repository contains the implementation correspondence, pinned Lean development, proof audit, and deterministic structural checks. The paper is theory-focused and makes no empirical performance claim.

ETHICS STATEMENT

This work uses no human subjects or private user data.

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A APPENDIX ROADMAP

This appendix gives the complete mathematical arguments and their implementation correspondence.

- **Implementation and objective.** Appendices B.1 and B.2 explain rollout-training mismatch and PPO clipping. Appendices E and E.1 give the population surrogate proof and supporting TV/KL identities.
- **Reward certificates and sampling.** Appendices C.2 to C.4 derive the random-length reduction and finite-sample reward guarantees. Appendices C.5 and C.6 prove the shared-baseline variance formula and the extension to data-dependent fitting in a pre-specified covered policy class. Appendix C.7 derives explicit certificate inputs from uniform logit-change bounds.
- **Advantage construction.** Appendices C.1 and C.8 give the RLOO and sparse-reward arguments. Appendices C.10 and C.11 analyze numerical safeguards and the optional uniform-scale branch.
- **Prefix filtering.** Appendices D.1 and D.3 prove the conditional masking comparisons and accepted-prefix mechanism bound. Appendix D.4 collects the additional concentration assumption, its consequences, and their proofs. Appendix D.5 covers alternative prefix gates.

B IMPLEMENTATION DIVERGENCE AND TRUST-REGION ANCHORING

B.1 DETAILS ON OFF-POLICY MISMATCH IN LLM-RL

Modern LLM-RL pipelines decouple inference from training: rollouts are produced by an inference engine (e.g., vLLM or SGLang) under the rollout policy π_{roll} , while gradient updates are computed in a separate training stack under the training policy π_{θ} . This architectural separation can introduce off-policy mismatch even when parameters are nominally aligned.

Mismatch Sources Common sources of mismatch include backend discrepancies (different attention kernels, precision formats, or reduction order), MoE routing discontinuities (Top- K routing can flip under small perturbations), and distributed staleness (actor-learner lag between rollout and consumption). These effects can compound autoregressively over long horizons and lead to volatile per-token ratios.

Anchoring and the Apparent Three-Distribution Structure Practical pipelines can exhibit an apparent three-distribution structure: (i) a rollout engine generates tokens according to an inference-time distribution, (ii) a training framework computes gradients under π_{θ} , and (iii) some implementations recompute the behavior log-probabilities inside the training stack, yielding a recomputed anchor that can differ from the true rollout distribution even when parameters are identical. Importantly, this third object is not an additional policy in our theory; it is an alternative implementation-time anchor for the denominator of ρ_t .

Trust Region Setup Qi et al. (2026) argue that the trust region (and hence the denominator of the importance ratio) must be defined with respect to the *original* behavior policy that generated the data, rather than a recomputed policy distribution evaluated inside the training stack. Accordingly, we denote this rollout-time behavior policy by π_{roll} , define $\rho_t := \pi_{\theta}(y_t | c_t) / \pi_{\text{roll}}(y_t | c_t)$, and assume access to rollout-time cached log-probabilities $\log \pi_{\text{roll}}(y_t | c_t)$. The theoretical development therefore involves only the policy pair $(\pi_{\text{roll}}, \pi_{\theta})$, consistent with long-horizon trust-region analyses for LLM-RL (e.g., Li et al. (2025)).

PPO-Style π_{old} Mismatch The implementation has a frozen PPO snapshot π_{old} in addition to the data-generating rollout policy π_{roll} . Its clipped PPO ratio and its vLLM correction ratio are respectively

$$r_t^{\theta/\pi_{\text{old}}} = \frac{\pi_{\theta}(y_t | c_t)}{\pi_{\text{old}}(y_t | c_t)}, \quad w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} = \frac{\pi_{\text{old}}(y_t | c_t)}{\pi_{\text{roll}}(y_t | c_t)}.$$

Their product is the behavior-to-current ratio $\rho_t^{\theta/\pi_{\text{roll}}} = r_t^{\theta/\pi_{\text{old}}} w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$. The prefix mask is computed from $w_t^{\pi_{\text{old}}/\pi_{\text{roll}}}$ and is detached, while gradients flow through the PPO ratio. The abstract prefix theorems apply to whichever policy pair supplies the ratio; applying them to the current policy during stale updates requires the finite-prefix coupling condition in ??.

B.2 WHY PPO CLIPPING CAN LEAK GRADIENTS UNDER MISMATCH

PPO Clipped Surrogate PPO (Schulman et al., 2017) constrains policy updates by clipping the current-to-snapshot ratio. In the notation of this paper, the ratio is

$$r_t^{\theta/\pi_{\text{old}}} := \frac{\pi_{\theta}(y_t | c_t)}{\pi_{\text{old}}(y_t | c_t)}.$$

The clipped surrogate takes the form

$$\mathcal{L}^{\text{CLIP}}(\pi_{\theta}) = \mathbb{E}_{\pi_{\text{roll}}} \left[\sum_{t=1}^T w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} \min \left(r_t^{\theta/\pi_{\text{old}}} \hat{A}_t, \text{clip}(r_t^{\theta/\pi_{\text{old}}}, 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) \hat{A}_t \right) \right],$$

where $\hat{A}_t$ is an advantage estimator and $\epsilon_{\text{clip}} > 0$ is the clipping threshold.

Asymmetry and Gradient Leakage The clipping operator is asymmetric in the sign of $\hat{A}_t$. When $\hat{A}_t < 0$ and $r_t^{\theta/\pi_{\text{old}}} \gg 1 + \epsilon_{\text{clip}}$, the unclipped term can still be selected:

$$r_t^{\theta/\pi_{\text{old}}} > 1 + \epsilon_{\text{clip}}, \hat{A}_t < 0 \implies r_t^{\theta/\pi_{\text{old}}} \hat{A}_t < (1 + \epsilon_{\text{clip}}) \hat{A}_t = \text{clip}(r_t^{\theta/\pi_{\text{old}}}, 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) \hat{A}_t,$$

hence $\min(\cdot)$ returns $r_t^{\theta/\pi_{\text{old}}} \hat{A}_t$. In this regime, the gradient magnitude still scales with $r_t^{\theta/\pi_{\text{old}}}$, so extremely large ratios can produce large updates even though they lie far outside the nominal trust region.

Why This Is Benign in Standard RL In classical on-policy settings, large ratios typically reflect genuine policy changes that over-emphasize low-quality actions; keeping the unclipped penalty for $\hat{A}_t < 0$ can be desirable as it discourages such actions.

Failure Mode Under Implementation Divergence In LLM-RL pipelines with inference–training mismatch, a large rollout-to-snapshot coefficient can arise from numerical or backend discrepancies rather than meaningful policy improvements. For example, a routing discontinuity in a Mixture-of-Experts model can yield a token v such that

$$\pi_{\text{roll}}(v | c_t) \approx 0.001, \quad \pi_{\text{old}}(v | c_t) \approx 0.9, \quad w_t^{\pi_{\text{old}}/\pi_{\text{roll}}} \approx 900.$$

Such spikes reflect implementation-level divergence and can multiply the PPO loss by a massive, erroneous detached coefficient. This motivates using a detached prefix trust-region mask to discard mismatched tokens rather than relying on PPO clipping, which acts on the separate current-to-snapshot ratio.

C REINFORCE MAX: SUPPLEMENTARY DETAILS

C.1 LEAVE-ONE-OUT BASELINE PROPERTIES

We summarize standard properties of the leave-one-out baseline estimator used in ??; see Ahmadian et al. (2024) for a detailed treatment.

Remark C.1 (Properties of the Leave-One-Out Baseline). Fix a prompt x and define its conditional reward objective by

$$J_x(\theta) := \mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} [R(x, y)].$$

Draw $n \geq 2$ i.i.d. responses $\{y^{(1)}, \dots, y^{(n)}\} \sim \pi_{\theta}(\cdot | x)$ with rewards $r_i = R(x, y^{(i)})$. Let

$$f_i = \nabla_{\theta} \log P^{\pi_{\theta}}(y^{(i)} | x)$$

denote the trajectory score function for sample i , and define the per-sample policy gradient contribution with baseline b as $g_i(b) = (r_i - b)f_i$. With the leave-one-out baseline $b_i = \frac{1}{n-1} \sum_{j \neq i} r_j$, the estimator

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g_i(b_i)$$

satisfies the following statements, provided the usual score-function regularity conditions hold: the n responses are conditionally i.i.d. given x ; r_i and f_i are integrable, f_i has a finite second moment, the products needed below are integrable (in particular $r_i f_i$ and $(r_i - \mu(x))^2 f_i f_i^\top$), and differentiation may be interchanged with the expectation defining $J_x(\theta)$. In particular, the normalized policy has support independent of θ (or obeys the corresponding boundary conditions), so that $\mathbb{E}[f_i | x] = 0$. Under these conditions:

1. **Conditional unbiasedness:** $\mathbb{E}[\hat{g} | x] = \nabla_\theta J_x(\theta)$.
2. **Conditional baseline independence:** For each i , b_i is independent of (r_i, f_i) given x ; in particular, $\mathbb{E}[b_i f_i | x] = 0$ and $\text{Cov}(f_i, b_i | x) = 0$. The moment assumptions ensure that these products and covariances are well-defined.
3. **Mean-baseline scaling:** Let $b^{\text{mean}} = \frac{1}{n} \sum_{j=1}^n r_j$ and $\hat{g}^{\text{mean}} = \frac{1}{n} \sum_{i=1}^n (r_i - b^{\text{mean}}) f_i$. Then deterministically $\hat{g}^{\text{mean}} = \frac{n-1}{n} \hat{g}$, and hence $\mathbb{E}[\hat{g}^{\text{mean}} | x] = \frac{n-1}{n} \nabla_\theta J_x(\theta)$.
4. **Second-moment decomposition:** Let $\mu(x) = \mathbb{E}[r_i | x]$ and $\sigma^2(x) = \text{Var}(r_i | x)$. Then $\text{Var}(b_i | x) = \sigma^2(x)/(n-1)$, and

$$\mathbb{E}[(r_i - b_i)^2 f_i f_i^\top | x] = \mathbb{E}[(r_i - \mu(x))^2 f_i f_i^\top | x] + \text{Var}(b_i | x) \mathbb{E}[f_i f_i^\top | x].$$

For a prompt distribution P independent of θ , the paper's global objective is $J(\pi_\theta) = \mathbb{E}_{x \sim P}[J_x(\theta)]$. If differentiation also commutes with this outer expectation, then

$$\mathbb{E}_{x, Y_{1:n}}[\hat{g}] = \mathbb{E}_{x \sim P}[\nabla_\theta J_x(\theta)] = \nabla_\theta J(\pi_\theta).$$

For a fixed finite prompt law, this follows by differentiating a finite weighted sum; the joint prompt/response sampling law and the complete RLOO derivative are checked by `joint_prompt_response_mass_normalized`, `joint_prompt_response_mass_nonneg` and `finite_prompt_rloo_policy_gradient` in `formal/PromptGradient.lean`. Fixing one prompt cannot substitute for this averaging step: two equally likely prompts with Bernoulli success probabilities θ and $1 - \theta$ have a constant global reward $1/2$, while the first prompt's reward derivative is 1. Their valid policies on $0 < \theta < 1$ and the derivative separation are checked by `two_prompt_policy_valid`, `two_prompt_reward` and `fixed_prompt_gradient_ne_global`.

The same statement does not require a finite prompt set. Let μ be an arbitrary prompt measure and interpret the global objective as the Bochner integral $\int J_x(\theta) d\mu(x)$. If the conditional policy regularity above holds μ -almost everywhere, the conditional RLOO differential is almost-everywhere measurable, and the prompt integrand is locally Lipschitz in θ with an integrable dominating bound (together with the corresponding measurability and integrability conditions), then differentiation under the prompt integral gives

$$DJ(\theta) = \int \mathbb{E}[\hat{g} | x] d\mu(x).$$

These are the standard dominated differentiation conditions; they make the outer exchange explicit rather than treating it as an automatic consequence of conditional unbiasedness. The finite-response RLOO calculation and this measure-level extension are checked by the following two Lean theorems:

```
ae_prompt_rloo_policy_gradient
measure_prompt_rloo_policy_gradient.
```

They are both defined in `formal/PromptGradientMeasure.lean`.

Remark C.2 (Relationship Between the Two Estimators). The deterministic proportionality $\hat{g}^{\text{mean}} = \frac{n-1}{n} \hat{g}$ implies that the mean-baseline estimator is a rescaled version of the leave-one-out estimator. Its smaller raw variance is entirely due to the constant scaling factor. In practice, this shrinkage can be absorbed into the learning rate, but it breaks exact unbiasedness whenever the target gradient is nonzero. The main advantage of the leave-one-out baseline is that b_i is independent of sample i given the prompt, which simplifies moment calculations and avoids self-correlation terms that arise when the baseline depends on r_i .

Remark C.3. The leave-one-out baseline is equivalent to the leave-one-out control variate from the classical REINFORCE literature (Williams, 1992). In the LLM-RL setting with n samples per prompt, each sample’s baseline is conditionally independent of its own trajectory given the prompt, making it particularly well-suited for group-based sampling strategies.

The stochastic statements in this remark are conditional on the stated independence, integrability, and differentiation assumptions. The finite Lean audit verifies the deterministic RLOO scaling, zero-sum, and same-sign degeneracy identities, as well as the independent product-space control-variate identity and scalar and cross-coordinate second-moment decompositions. The finite i.i.d. leave-one-out construction is checked directly, including $\text{Var}(b_i) = \sigma^2/(n-1)$ and every entry of the displayed outer-product identity. Extensions to general probability spaces and conditional expectations retain the stated regularity assumptions. One finite product-space building block is now checked explicitly: if p is the trajectory distribution, u is an independent baseline-sample distribution, and the score f has zero p -mean, then

$$\sum_{i,j} p_i u_j (r_i - b_j) f_i = \sum_i p_i r_i f_i.$$

This is the finite form of the control-variate step $\mathbb{E}[b_i f_i] = \mathbb{E}[b_i] \mathbb{E}[f_i] = 0$ and is verified by `finite_independent_baseline_score` in `formal/RLOOProbability.lean`. The theorem does not by itself discharge the general probability-space and differentiation-expectation obligations. As concrete finite checks, `finite_iid_rloo_unbiased_two` proves the two-sample leave-one-out estimator exactly, while `finite_iid_rloo_unbiased` proves the arbitrary finite group-size case directly: for $m+1$ independent draws from a finite mass function p (with $m \neq 0$), the average leave-one-out estimator using the other m rewards has expectation $\sum_y p(y) r(y) f(y)$ whenever $\sum_y p(y) f(y) = 0$. The abstract interface `finite_rloo_independent_baselines` remains useful when baseline laws are specified independently of this concrete i.i.d. construction. The finite-dimensional score version is checked componentwise by `finite_iid_rloo_unbiased_vector`; hence the displayed vector estimator’s expectation equals the target score expectation on finite response support. The derivative connection is now also checked on finite response support. Let $p_\theta(y)$ be locally nonnegative and normalized, differentiable at θ , and let $R(y)$ be independent of θ . Define its score differential as $s_\theta(y) = p_\theta(y)^{-1} D p_\theta(y)$ on positive-mass responses and as zero otherwise. Local nonnegativity implies $D p_\theta(y) = 0$ at a zero-mass response. Thus

$$\sum_y p_\theta(y) s_\theta(y) = 0, \quad D \left(\sum_y p_\theta(y) R(y) \right) = \sum_y p_\theta(y) R(y) s_\theta(y) = \mathbb{E}[\hat{g}].$$

In a finite-dimensional Euclidean parameter space this differential is identified with the usual gradient. The certificate `finite_iid_rloo_policy_gradient` in `formal/PolicyGradient.lean` proves these identities from local normalization, nonnegativity, and differentiability, rather than assuming a zero-mean score or an exchange of derivative and finite expectation. On positive-mass responses, `policyScore_is_log_derivative` identifies $s_\theta(y)$ with $D \log p_\theta(y)$. The general probability-space statement above retains the listed integrability and differentiation assumptions. This certificate concerns the raw on-policy RLOO estimator; the changes introduced by normalization, prefix gates, off-policy weights, and clipping are treated separately in ???. The connection to the paper’s autoregressive model is explicit: `tree_iid_rloo_policy_gradient` in `formal/TreePolicyGradient.lean` constructs the response masses by multiplying history-dependent token probabilities and proves the result for the recursively defined terminal-reward objective used in the performance-difference theorems. Differentiability of the response masses follows from differentiability of the conditional

token probabilities. This permits common absorbing EOS transitions and makes no independence assumption across positions within a response.

For a scalar score coordinate, the corresponding finite second-moment calculation is also checked. If p is the trajectory mass, u is an independent baseline mass with mean μ , then

$$\sum_{i,j} p_i u_j (r_i - b_j)^2 f_i^2 = \sum_i p_i (r_i - \mu)^2 f_i^2 + \left(\sum_j u_j (b_j - \mu)^2 \right) \left(\sum_i p_i f_i^2 \right).$$

This is the scalar finite product-space form of the conditional second-moment decomposition in the statement above and is verified by `finite_independent_baseline_second_moment`. The cross-coordinate version is verified by `finite_independent_baseline_cross_moment` and gives each entry of a finite-dimensional outer-product decomposition. In `formal/RLOOMoments.lean`, `finite_loo_baseline_marginal` derives the baseline law by separating sample i from the other $n - 1$ draws in the actual i.i.d. product mass. From this construction, `finite_loo_baseline_variance` proves $\text{Var}(b_i) = \sigma^2/(n - 1)$, and `finite_loo_score_cross_moment` proves

$$\mathbb{E}[(r_i - b_i)^2 f_{i,k} f_{i,l}] = \mathbb{E}[(r_i - \mu)^2 f_{i,k} f_{i,l}] + \frac{\sigma^2}{n - 1} \mathbb{E}[f_{i,k} f_{i,l}]$$

for any two score coordinates k, l . No independent-baseline variance formula is assumed in this final certificate. This is a per-sample second-moment identity; the covariance of the group-averaged estimator also involves cross-sample terms because its leave-one-out baselines share samples. The exact scalar group-average variance is now given in ?? and Appendix C.5. Its finite-law identity applies to any zero-mean scalar signal, including each score coordinate here and the trajectory ratio residual used in the reward certificate.

The finite proofs are supplied in the accompanying Lean files; see also Ahmadian et al. (2024) for background.

C.2 AUTOREGRESSIVE RLOO POPULATION BRIDGE AND RANDOM REDUCTION

The finite-tree formalization also connects the estimator actually used by RLOO to the population surrogate in the performance-difference analysis. Let p be the rollout tree policy, q a candidate policy, and let $\rho_t(y)$ be the candidate-to-rollout token ratio at the t th history on a finite horizon. The rollout trajectory residual used in the formalization is $h(y) = \sum_t (\rho_t(y) - 1)$ and has zero p -mean. Consequently, for a group of $m + 1$ i.i.d. trajectories, the concrete leave-one-out estimator satisfies

$$\mathbb{E}_p \left[(m + 1) \hat{\mathcal{L}}_{\text{RLOO}} \right] = (m + 1) \mathcal{S}_p(q),$$

and its group-size-normalized expectation is exactly the raw population surrogate $\mathcal{S}_p(q)$. This allows arbitrary history dependence across response tokens; only the trajectories in the group are i.i.d. It is proved by `response_ratio_residual_mean_zero`, `iid_rloo_surrogate_increment_expectation`, and `group_surrogate_ratio` in `formal/RLOOSurrogate.lean`.

The bridge concerns the raw, unclipped surrogate. Prefix gating, adaptive normalization, off-policy weighting, and PPO clipping remain the explicit correction terms in ??; they are not silently absorbed into an unbiasedness claim. The reduction itself must also be stated precisely. If a response contributes Z over $L > 0$ active tokens, response-level averaging estimates $\mathbb{E}[Z/L]$, while global token-level averaging estimates $\kappa = \mathbb{E}[Z]/\mathbb{E}[L]$. The exact bias decomposition is

$$\mathbb{E}[Z/L] - \kappa = - \frac{\mathbb{E}[(L - \mathbb{E}[L])Z/L]}{\mathbb{E}[L]}.$$

For the independent-block analysis, Z and L instead denote full-group sums. A group response mean remains the mean of its individual Z_i/L_i , not the ratio of group sums: for contributions $(1, -1)$ and lengths $(1, 3)$, these quantities are $1/3$ and 0 , respectively. This arithmetic distinction is checked by `group_ratio_ne_response_mean` in `formal/BatchReduction.lean`. For i.i.d. blocks with $\bar{L} \geq \ell > 0$, the self-normalized token estimator has the finite bounds

$$\mathbb{E}[(\bar{Z}_n/\bar{L}_n - \kappa)^2] \leq \frac{\mathbb{E}[(Z - \kappa L)^2]}{n\ell^2}, \quad \Pr(|\bar{Z}_n/\bar{L}_n - \kappa| \geq \varepsilon) \leq \frac{\mathbb{E}[(Z - \kappa L)^2]}{n\ell^2\varepsilon^2}.$$

These are checked by `ratio_expectation_gap`, `iid_token_mean_mse_of_ratio`, and `iid_token_mean_tail` in `formal/BatchReduction.lean`.

The finite performance certificate in `formal/BatchPerformance.lean` composes this estimation error with a valid population lower bound and explicitly states that it is for a fixed candidate policy evaluated on independent blocks; it is not a uniform guarantee after selecting the candidate from the same data.

C.3 PROOF OF THE PROMAX PERFORMANCE CERTIFICATE

Fix the candidate family before drawing evaluation blocks. A block is the pair $(x, (y_1, \dots, y_n))$, with mass $P(x) \prod_{i=1}^n P^p(y_i | x)$. The finite history tree constructs each trajectory mass from its conditional token kernels; summing over responses and prompts gives a normalized block law. This is checked by `block_mass_normalized`. Conditional on x , the response residual $h_j(y) = \sum_t (\rho_{j,t}(y) - 1)$ has zero rollout mean. Each b_i is a function of the other $n - 1$ responses and is independent of y_i given x . Therefore

$$\mathbb{E}[Z_j | x] = n \mathbb{E}_{y \sim p(\cdot | x)}[R(x, y) h_j(y)] = n S_{j,x}.$$

The equalities use the actual leave-one-out baseline; shared baselines are not replaced by independent copies. Prompt averaging proves $\mathbb{E}[Z_j] = n S_j$, while $\mathbb{E}[L] = n \mu_L$. These steps are checked by `block_increment_expectation`, `block_length_expectation`, and `block_ratio_target`.

For the full objective, let $D = \sum_b L_b$ and use weights $m_{b,i,t}/D$. At the comparison policy p , the frozen-snapshot PPO ratio is $r_0 = p/\pi_{\text{old}}$ and satisfies $r_0 w = 1$. At q_j it is $r_j = q_j/\pi_{\text{old}}$ and satisfies $r_j w = \rho_j$. Subtracting the two exact objective decompositions gives

$$\begin{aligned} \Delta_j S_{\text{PM}} + \Delta_j G - \Delta_j N + \Delta_j C_{\text{clip}} &= \frac{1}{D} \sum_{b,i,t} m_{b,i,t} A_{b,i} (\rho_{j,b,i,t} - 1) \\ &= \frac{\sum_b Z_{j,b}}{\sum_b L_b}. \end{aligned}$$

The second equality uses the zero ratio residual on shared absorbing padding. The gate, normalized signal, snapshot coefficients and clipping terms remain in the first expression without independence requirements. `response_residual_eq_token_sum` and `group_increment_eq_token_sum` connect the token indices to the history-tree ratio trace; `corrected_batch_change_eq_rloo_ratio` and `corrected_block_objective_eq` verify the complete conversion. Configured uniform-scale signals are included in the actual normalized signal, so their difference from raw RLOO is retained in N .

Write $F_j = Z_j - \kappa_j L$. The population identities give $\mathbb{E}[F_j] = 0$. The blocks are independent, so $\mathbb{E}[\bar{F}_j^2] = v_j/K$, even though siblings within a block are dependent. Since $\bar{L} \geq n\ell$, the mean-square error of $\hat{\kappa}_j$ is at most $v_j/(K(n\ell)^2)$. Markov's inequality on its square gives the individual tail bound, and a finite union bound gives Eq. equation ???. No independence between candidate estimates is needed. `rloo_uniform_estimation_bound` verifies this result with the block law and its population means constructed above.

Finally, Theorem 10.10 yields $J(q_j) - J(p) \geq S_j - B_j = \mu_L \kappa_j - B_j$. On the simultaneous estimation event, $\kappa_j \geq \hat{\kappa}_j - \eta_j$ for every candidate, proving the performance bound. If the selected candidate has nonpositive true gain but passes the strict threshold, its estimation error must exceed its margin. That event is a subset of the union event already bounded above. `rloo_performance_from_estimate` and `rloo_selected_false_improvement_bound` check these conclusions, including the actual Adaptive cost and true reward values on the same finite prompt/history model.

C.4 PROOF OF THE OBSERVED-LENGTH AND DRIFT-ENERGY CERTIFICATE

For one prompt, define $h(y) = \sum_{t=1}^{|y|} (\rho_t(y) - 1)$ and $e(c) = \sum_a p(a | c)(q(a | c)/p(a | c) - 1)^2$. Normalization and $q \ll p$ imply

$$\sum_a p(a | c)(q(a | c)/p(a | c) - 1) = 0.$$

Zero-probability actions are omitted, or assigned zero weighted contribution. Hence for every real u ,

$$\sum_a p(a \mid c) (u + q(a \mid c)/p(a \mid c) - 1)^2 = u^2 + e(c).$$

Induction over continuation length on the normalized history tree yields

$$\mathbb{E}_p[h(Y)^2 \mid \text{prefix } c] = h(c)^2 + \mathbb{E}_p \left[\sum_{t > |c|} e(c_t) \mid \text{prefix } c \right].$$

The formal recursion is defined from each prefix kernel, including unreachable prefixes, with-out conditioning on a zero-probability event. At the root and after prompt averaging this gives $\mathbb{E}[H_j(Y)^2] = \mathcal{E}_j$. The Lean theorems `local_ratio_residual_mean_zero`, `residual_square_step`, `trajectory_residual_second_moment`, and `response_residual_energy` verify these steps. `ratio_energy_eq_prefix_sum` identifies the recursive cost with the explicit sum of prefix-law expectations. No token independence is used.

Within an actual group, each leave-one-out baseline lies in $[-R_{\max}, R_{\max}]$. Thus $|A_i| \leq 2R_{\max}$ and the finite Cauchy-Schwarz inequality gives

$$\left(\frac{Z_j}{n} \right)^2 \leq \frac{1}{n} \sum_{i=1}^n A_i^2 H_j(y_i)^2 \leq 4R_{\max}^2 \frac{1}{n} \sum_{i=1}^n H_j(y_i)^2.$$

Expectation under the constructed conditional-i.i.d. group law proves $\mathbb{E}[(Z_j/n)^2] \leq 4R_{\max}^2 \mathcal{E}_j$. This inequality uses the actual shared-baseline statistic; it does not factor dependent sibling contributions. It is verified by `rloo_estimator_square_bound`, `rloo_estimator_second_moment_bound`, and `block_surrogate_second_moment`.

The common batch denominator cancels exactly:

$$\frac{D}{Kn} \left(\frac{\sum_b Z_{j,b}}{D} \right) = \frac{1}{K} \sum_b \frac{Z_{j,b}}{n}.$$

The right-hand side averages unbiased group contributions, independently across the K blocks, even when reward, length, and gate decisions are dependent within blocks. Its variance is at most $4R_{\max}^2 \mathcal{E}_j / K$. `corrected_objective_rescaling` combines the full ProMax objective with this cancellation, and `rescaled_surrogate_unbiased` proves its expectation using the actual block law. `rescaled_surrogate_uniform_tail`, `rescaled_performance_from_estimate`, and `rescaled_selected_false_improvement_bound` complete the simultaneous confidence, performance, and selection guarantees.

Finally, a conditional energy bound $e(c) \leq d_j^2$ implies $\mathcal{E}_j \leq T d_j^2$ by summing prefix expectations. A ratio bound $|q_j/p - 1| \leq d_j$ need only hold on positive- p actions; shared zero actions impose no artificial requirement $d_j \geq 1$. `local_ratio_energy_of_bound` and `family_ratio_energy_of_ratio_bound` prove this implication. After absorbing EOS, both kernels put mass one on EOS and the energy is zero. The resulting square-root horizon scaling concerns the statistical margin only, and does not remove the separate policy-distribution-shift cost from the reward bound.

C.5 EXACT SHARED-BASELINE GROUP VARIANCE

Fix a prompt and suppress its index. Write $X_i = R_i - \mathbb{E}[R_i]$, $Y_i = H_j(y_i)$, $\theta = \mathbb{E}[XY]$, $a = \mathbb{E}[X^2]$, $b = \mathbb{E}[Y^2]$, and $c = \mathbb{E}[X^2 Y^2]$. Both X and Y have mean zero. Let $S_X = \sum_i X_i$, $S_Y = \sum_i Y_i$, and $S_{XY} = \sum_i X_i Y_i$. For every realized group, expanding the leave-one-out sums gives

$$U = \frac{n S_{XY} - S_X S_Y}{n(n-1)}.$$

Reward translation leaves the RLOO statistic unchanged, so this equality uses the actual reward estimator. These statements are checked by `rloo_eq_sample_covariance` and `rloo_reward_translation`.

Direct calculation under the finite i.i.d. group law gives

$$\begin{aligned}\mathbb{E}[S_{XY}^2] &= nc + n(n-1)\theta^2, \\ \mathbb{E}[S_{XY}S_XS_Y] &= nc + n(n-1)\theta^2, \\ \mathbb{E}[S_X^2S_Y^2] &= nc + n(n-1)(ab + 2\theta^2).\end{aligned}$$

For example, add one independent pair (X, Y) to the running sums. Conditioning on the previous sums, expanding the fourth-degree polynomial, and using $\mathbb{E}[X] = \mathbb{E}[Y] = 0$ leaves the recurrence

$$\mathbb{E}[S_X^2S_Y^2]_{n+1} - \mathbb{E}[S_X^2S_Y^2]_n = c + 2nab + 4n\theta^2.$$

The mixed third-sum recurrence is $c + 2n\theta^2$. The finite product mass is decomposed exactly using `iid_expect_cons`, `iid_sample_sum_square`, `iid_sum_product_cross`, and `iid_centered_sums_mixed_square` prove the displayed moments by induction. In particular, the terms from overlapping sample indices are included, rather than discarded by an independence assumption.

Since $\mathbb{E}[U] = \theta$, expansion of $\mathbb{E}[(nS_{XY} - S_XS_Y)^2]/[n^2(n-1)^2] - \theta^2$ yields

$$\text{Var}(U) = \frac{c - \theta^2}{n} + \frac{ab + \theta^2}{n(n-1)}.$$

This is verified by `centered_rloo_variance` and `actual_rloo_variance`. The first coefficient is $\text{Var}(XY)$ and the second is nonnegative; their signs are checked by `covariance_variance_coefficients_nonneg`. Under $|R| \leq R_{\max}$, one has $|X| \leq 2R_{\max}$, $a \leq R_{\max}^2$, and $c \leq 4R_{\max}^2b$. The coefficient of θ^2 after collecting terms is $(2-n)/[n(n-1)] \leq 0$. Therefore

$$\text{Var}(U) \leq \left(\frac{4}{n} + \frac{1}{n(n-1)} \right) R_{\max}^2b.$$

`actual_rloo_variance_bound` and `rloo_variance_factor_rate` verify the bound and $c_n \leq 5/n$.

For the autoregressive model, the conditional covariance target is exactly the prompt surrogate: $\theta_{j,x} = \mathcal{S}_{j,x}$, as verified by `response_centered_covariance`. The same history-tree ratio energy equals $b = \mathcal{E}_{j,x}$. Thus `response_rloo_variance_exact` and `response_rloo_variance_bound` instantiate the generic identity and bound on actual response groups. Expanding $U_j - \mathcal{S}_j = (U_j - \mathcal{S}_{j,x}) + (\mathcal{S}_{j,x} - \mathcal{S}_j)$ makes the cross term vanish under the conditional block law, giving

$$\text{Var}(U_j) = \tau_j^2 + \mathbb{E}_x[V_{j,x}(n)].$$

`block_surrogate_variance_exact` proves this equality directly from the prompt/response product mass. `group_surrogate_mse_exact` then proves the exact variance divided by K for the full empirical estimator; `group_surrogate_uniform_tail` and `group_selected_false_improvement_bound` supply the sharper confidence and selection guarantees. The bound taking the minimum with the earlier energy-only estimate is checked by `family_rloo_variance_combined_bound`.

C.6 POLICY FITTING WITHIN A FIXED COVER

Fix the policy class, centers and assignment map in `??`. These objects may be constructed conditionally on independent training data, but not chosen after observing the evaluation blocks. Write $j = \nu(\theta)$ and let U_θ denote the actual RLOO group mean. For every realized group,

$$\begin{aligned}|U_\theta - U_{\theta_j}| &\leq \frac{1}{n} \sum_i |R_i - b_i| |H_\theta(x, y_i) - H_{\theta_j}(x, y_i)| \\ &\leq 2R_{\max}\zeta.\end{aligned}$$

The leave-one-out baseline remains the mean of the other responses in this same group. No independence among the n summands is used in this deterministic bound. Averaging the K groups preserves the bound. On the population side, the previously proved RLOO/tree identity gives

$$|\mathcal{S}_\theta - \mathcal{S}_{\theta_j}| = |\mathbb{E}_p[R(H_\theta - H_{\theta_j})]| \leq R_{\max}\zeta.$$

Consequently, for every sample batch and every class member,

$$|\hat{\mathcal{S}}_\theta - \mathcal{S}_\theta| \leq |\hat{\mathcal{S}}_{\theta_j} - \mathcal{S}_{\theta_j}| + 3R_{\max}\zeta.$$

The event that any member exceeds its covered margin is therefore a subset of the finite union of center deviation events. Applying the exact group variance bound proves Eq. equation ???. If a fitted policy satisfies the positive-improvement threshold but has nonpositive true gain, the Adaptive lower bound forces it into that same deviation event. This proves the selection claim for any function of the evaluation data taking values in the fixed class, including continuous parameter fitting. The finite sampling space makes this event well-defined even for an infinite class index set.

The Lean proof in `formal/PolicyCover.lean` verifies the RLOO perturbation bound, population comparison, uniform event transfer, and reward lower bound. It also derives the sufficient radius Td from the finite token sum; the response-level cover is not inferred from an empirical check on the observed batch. The connection to the implemented objective is the same pointwise rescaling identity used in ??; applying it to a fitted policy does not require that policy to be independent of the batch.

C.7 CERTIFICATE CALIBRATION FROM NORMALIZED WEIGHTS AND LOGITS

We prove ?? using finite normalized-weight inequalities. They supply sufficient numerical bounds for the existing certificate; the general reward and cover theorems retain their original conditions.

Normalized-weight energy. Let p be a finite probability mass function, let $0 < \ell \leq v_a \leq u$, and write $Z = \sum_a p_a v_a$, $q_a = p_a v_a / Z$. Then $\ell \leq Z \leq u$ and q is a probability mass function with exactly the support of p . Actions of zero p -mass contribute zero to all expectations below. The pointwise inequality $(v_a - \ell)(u - v_a) \geq 0$ implies

$$\mathbb{E}_p[v^2] \leq (\ell + u)Z - \ell u.$$

Since $((\ell + u)Z - 2\ell u)^2 \geq 0$, the right-hand side is at most $(\ell + u)^2 Z^2 / (4\ell u)$. Therefore

$$\chi^2(q||p) = \mathbb{E}_p[(v/Z - 1)^2] = \frac{\mathbb{E}_p[v^2]}{Z^2} - 1 \leq \frac{(u - \ell)^2}{4\ell u} =: C(\ell, u). \quad (7)$$

This energy bound is sharp: put probability $u/(\ell + u)$ on weight ℓ and probability $\ell/(\ell + u)$ on weight u . Then $Z = 2\ell u/(\ell + u)$ and equality holds.

Reverse KL and TV. Dividing the pointwise quadratic inequality by $v_a > 0$ and averaging gives $Z + \ell u \mathbb{E}_p[1/v] \leq \ell + u$. The nonnegative square $(Z - \ell u \mathbb{E}_p[1/v])^2$ now yields

$$Z \mathbb{E}_p[1/v] \leq \frac{(\ell + u)^2}{4\ell u}.$$

Using $\log x \leq x - 1$ for $x > 0$, we obtain

$$D_{\text{KL}}(p||q) = \mathbb{E}_p[\log(Z/v)] \leq Z \mathbb{E}_p[1/v] - 1 \leq C(\ell, u).$$

Finally, weighted Cauchy–Schwarz gives

$$\text{TV}(p, q) = \frac{1}{2} \mathbb{E}_p[|v/Z - 1|] \leq \frac{1}{2} \sqrt{\mathbb{E}_p[(v/Z - 1)^2]} \leq \frac{1}{2} \sqrt{C(\ell, u)}.$$

The sharpness claim above concerns the energy bound, not equality in these additional KL and TV inequalities.

Softmax and common shifts. For $p_a = e^{z_a} / \sum_b e^{z_b}$ and $q_a = e^{z'_a} / \sum_b e^{z'_b}$, direct cancellation gives

$$q_a = \frac{p_a e^{z'_a - z_a - c}}{\sum_b p_b e^{z'_b - z_b - c}} \quad \text{for every common shift } c \in \mathbb{R}.$$

If the logit increments lie in an interval $[c, c + \omega]$, their centered exponentials lie in $[1, e^\omega]$. Thus $C(1, e^\omega) = (e^\omega - 1)^2 / (4e^\omega)$. The shift may depend on the context. This argument retains the full softmax normalizer, so the bound is independent of the smallest positive token probability. A shared masked support can be treated as the finite action set at that row; weights on shared zero-probability actions may be set to one in the normalized-tilt formulation. The argument does not permit a change in sampling support.

History trees and the corrected ProMax statistic. Apply the local bounds at every prompt and history of length less than T . The additive energy identity gives $\mathcal{E}_j \leq TC_j$, and averaging local TV under any rollout occupancy preserves the bound $\epsilon_j = \sqrt{C_j}/2$. Substitution in the already proved Adaptive bound gives $B_j \leq \bar{B}_j$ from Eq. equation ???. Finite prompt averaging preserves both inequalities. Shared absorbing EOS transitions have zero local energy and divergence; the uniform horizon bound may overestimate their contribution.

For the actual RLOO group estimator, the previously proved second-moment bound becomes $\mathbb{E}[(Z_j/n)^2] \leq 4R_{\max}^2 TC_j$. Independence of complete groups, Chebyshev’s inequality, and a finite union bound give Eq. equation ???. The exact observed-token rescaling still identifies this estimator with $D(\Delta_j S_{\text{PM}} + \Delta_j G - \Delta_j N + \Delta_j C_{\text{clip}})/(Kn)$. Combining its deviation bound with the population performance lower bound proves the stated reward certificate. All gate, normalization, clipping, random-length, and shared-baseline terms therefore retain their earlier definitions. A sharper valid group-variance bound can replace this second-moment estimate through ??.

A sufficient logit-cover radius. Let a center q' be a normalized tilt of rollout policy p with weights in $[\ell_0, u_0]$, and let a class member q be a normalized tilt of q' with weights in $[\ell_1, u_1]$. At each context, $q'_a/p_a \leq u_0/\ell_0$ on positive- p actions, and $|v_a/\mathbb{E}_{q'}[v] - 1| \leq (u_1 - \ell_1)/\ell_1$. Consequently,

$$\left| \frac{q_a}{p_a} - \frac{q'_a}{p_a} \right| \leq \frac{u_0}{\ell_0} \frac{u_1 - \ell_1}{\ell_1}.$$

At shared zero-probability actions both totalized ratios are zero. The triangle inequality over T positions bounds the difference of the additive response residuals by T times this quantity. Taking $(\ell_0, u_0) = (1, e^\omega)$ and $(\ell_1, u_1) = (1, e^r)$ proves the radius $Te^\omega(e^r - 1)$. This provides a sufficient metric for the fixed cover in ??; it does not establish how many centers suffice for a given neural policy class.

`formal/LogitBounds.lean` verifies the normalized-weight inequalities, the energy equality example, softmax normalization and shift invariance, the token-to-response cover bound, and the finite-prompt composition with the actual RLOO sampling and reward certificate.

C.8 OBJECTIVE CONSISTENCY OF $\gamma = 1.0$

Remark C.4 (On Setting $\gamma = 1.0$ for Sparse Terminal Rewards). For a response with T active tokens and shaped reward $\tilde{r}$ assigned only at the final token, the reward-to-go recurrence $G_t = r_t + \gamma G_{t+1}$ gives

$$G_t = \gamma^{T-t} \tilde{r}, \quad 1 \leq t \leq T.$$

Thus $\gamma = 1$ gives $G_t = \tilde{r}$ at every active position, as required by the undiscounted sequence-reward objective $J(\pi_\theta) = \mathbb{E}[R(x, y)]$ and the token expansion in ??. For nonzero $\tilde{r}$, requiring this equality at every suffix length also forces $\gamma = 1$. Both claims are checked by `sparse_terminal_return_gamma_one` and `gamma_one_iff_constant_sparse_return` in `formal/SparseReturn.lean`. Choosing $0 \leq \gamma < 1$ introduces position-dependent factors γ^{T-t} and downweights earlier token contributions. Such an estimator generally no longer targets the undiscounted policy gradient; identifying it with a discounted-objective gradient additionally requires specifying the outer time weights in the policy-gradient formula. See Williams (1992) for background on REINFORCE.

C.9 THE NORMALIZATION MECHANISM FOR ACTUAL RLOO SIGNALS

Proof of ??. Let the group contain k rewards r_+ and m rewards r_- , with $n = k + m \geq 2$ and $\Delta = r_+ - r_- > 0$. The reward sum is $kr_+ + mr_-$. Subtracting the other-response mean gives

$$r_+ - \frac{(k-1)r_+ + mr_-}{n-1} = \frac{m\Delta}{n-1} = a_+, \quad r_- - \frac{kr_+ + (m-1)r_-}{n-1} = -\frac{k\Delta}{n-1} = -a_-.$$

Both magnitudes are strictly positive. Repeating each response advantage at its active tokens gives positive and negative first-moment magnitudes Pa_+ and Na_- , squared sums Pa_+^2 and Na_-^2 , and nonzero count $P + N$. Therefore

$$\kappa = \frac{Pa_+}{Na_-}, \quad D = Pa_+^2 + \kappa^2 Na_-^2 = a_+^2 \frac{P(P+N)}{N}.$$

Since $\alpha = \sqrt{(P + N)/D}$ and $\beta = \alpha\kappa$,

$$\alpha a_+ = \sqrt{N/P}, \quad \beta a_- = \sqrt{P/N}.$$

Substituting the RLOO magnitudes also gives

$$\kappa = \frac{Pm}{Nk} = \frac{P/k}{N/m} = \frac{\bar{T}_+}{\bar{T}_-}.$$

Multiplication by the token counts gives equal aggregate coefficient magnitudes $P\sqrt{N/P} = N\sqrt{P/N}$. All steps use the actual reward group and its response lengths. $\square$

The finite construction in `formal/AdaptiveMechanism.lean` indexes responses by the disjoint union of the higher- and lower-reward groups. It computes the reward sum and the RLOO signals, then forms the positive, negative and squared sums using arbitrary finite response-length weights. `finite_binary_rloo_normalization` composes these steps with the normalization map. The ratio and mass-balance identities are checked in the same module. The older equal-amplitude algebra remains a special case; it is not used to assume that binary RLOO advantages are ± 1 .

What is balanced. Before normalization, the ratio of positive to negative absolute token coefficient mass is

$$\frac{Pa_+}{Na_-} = \frac{\bar{T}_+}{\bar{T}_-}.$$

Thus unequal average lengths directly produce the scalar mass imbalance; class frequency alone does not, because RLOO already assigns different magnitudes to the two classes. After normalization, the two total coefficient magnitudes are equal. At the token level, $P \ll N$ gives $|\hat{A}_+| \gg |\hat{A}_-|$, but this is not the assertion $\beta < 1$. For one success and nine equally long failures with rewards 1 and 0, $\alpha = \beta = 3$; the negative amplitude grows from $1/9$ to $1/3$. The Lean counterexample `rare_success_does_not_imply_negative_attenuation` checks these factors. Importance weights, gates, and score vectors can change the final update, so coefficient balance is not a gradient-vector balance theorem.

Proof of ??. For positive c , scaling A to cA preserves the two sign sets and the nonzero-token count. The positive and negative first sums both acquire a factor c , and both squared sums acquire a factor c^2 . Consequently, κ is unchanged, D becomes c^2D , and the ideal factors become α/c and β/c . Their products with cA equal the original normalized advantages at every active position. The same calculation holds with arbitrary fixed nonnegative token or response-length weights.

For a group of size $n \geq 2$, direct rearrangement gives

$$r_i - \frac{1}{n} \sum_j r_j = \frac{n-1}{n} \left(r_i - \frac{\sum_j r_j - r_i}{n-1} \right).$$

Apply scale invariance with $c = (n-1)/n > 0$ to prove the baseline equivalence. $\square$

The same Lean module proves all five finite weighted-moment transformations, invariance of the resulting normalization map, and the equality obtained from the two actual baseline formulas. This equivalence assumes the same ideal normalization; it does not compare different normalization rules or different prompt groups. The independent-baseline identities and group-variance bounds elsewhere in the paper apply to the raw RLOO surrogate before this data-dependent map.

Proof of ??. Write $a = a_+ > 0$ and $b = a_- > 0$. The token-weighted mean and population variance are

$$\mu = \frac{Pa - Nb}{P + N}, \quad v = \frac{P(a - \mu)^2 + N(-b - \mu)^2}{P + N} = \frac{PN(a + b)^2}{(P + N)^2} > 0.$$

The centered positive and negative coefficients are $a - \mu = N(a + b)/(P + N)$ and $-b - \mu = -P(a + b)/(P + N)$. Dividing by $\sqrt{v}$ gives $\sqrt{N/P}$ and $-\sqrt{P/N}$, respectively, which are the asymmetric coefficients in ?? . These variance and square-root identities are checked by `two_level_token_zscore`. $\square$

A sign change with actual multivalued RLOO rewards. Consider three responses with rewards $(0, 2/3, 1)$ and active-token lengths $(1, 1, 4)$. Their RLOO advantages and token-weighted moments are

$$(A_1, A_2, A_3) = (-5/6, 1/6, 2/3), \quad \mu_{\text{tok}} = 1/3, \quad v_{\text{tok}} = 11/36.$$

Although $A_2 = 1/6 > 0$, its centered value is $A_2 - \mu_{\text{tok}} = -1/6$, so its ordinary token-standardized coefficient is negative. The ideal asymmetric coefficient is $\alpha A_2 > 0$ because $\alpha > 0$. Both sign classes are present, and every reward lies in $[0, 1]$; the example uses the actual leave-one-out formula and includes no zero-advantage tokens. The finite RLOO values, weighted mean and variance, negative standardized coefficient, and positive asymmetric coefficient are checked in `formal/AdaptiveMechanism.lean`. This illustrates a distinction for multivalued rewards, consistent with the exact binary equivalence in ??.

Connection to the numerical implementation. If the normalization branch is taken and the product and factor clamps are inactive, the implementation replaces D by $D + \varepsilon_{\text{num}}$. The factor identity

$$\sqrt{\frac{P+N}{D+\varepsilon_{\text{num}}}} = \sqrt{\frac{D}{D+\varepsilon_{\text{num}}}} \sqrt{\frac{P+N}{D}}$$

shows that the two normalized amplitudes have the same multiplicative shrinkage; their relative factor ratio and zero-mean balance are preserved. This identity is checked by `stabilizer_common_shrinkage`. Under a reward rescaling, however, D changes while the fixed numerical stabilizer does not, so exact reward-scale invariance need not survive. Clamps and fallbacks have the separate explicit formulas in Proposition C.6.

C.10 NUMERICAL STABILITY

The adaptive normalization (??) includes several safeguards:

Remark C.5 (Numerical Thresholds vs. Divergence Bounds). The symbols ε_{num} and ε_{tol} in this appendix denote implementation hyperparameters used for numerical stability and uniform-reward detection. They are unrelated to the divergence upper bound ϵ in Definition 10.2.

1. **Sum threshold:** If $\sum_{(i,t) \in \mathcal{P}} A_{i,t} < \varepsilon_{\text{num}}$ or $-\sum_{(i,t) \in \mathcal{N}} A_{i,t} < \varepsilon_{\text{num}}$ (default $\varepsilon_{\text{num}} = 10^{-8}$), normalization is skipped and the original advantages are returned.
2. **Additive stabilizer:** The denominator inside the square root for α always includes $\varepsilon_{\text{num}} > 0$. This changes the second moment even when every clamp is inactive.
3. **Scale clamping:** α and β are clamped to $[\varepsilon_{\text{num}}, s_{\text{max}}]$ (default $s_{\text{max}} = 10$) to prevent extreme scaling.
4. **Intermediate overflow protection:** The quantity $\kappa^2 \sum_{(i,t) \in \mathcal{N}} A_{i,t}^2$ (where κ is defined in ??) is clamped to 10^8 before computing α .
5. **Finiteness check:** If α or β is NaN or $\pm\infty$ after computation, normalization is skipped.

The following proposition quantifies how these safeguards change the ideal empirical constraints. A small deviation requires corresponding bounds on the stabilizer and clamp residuals; it is not automatic from the presence of these safeguards. The defaults remain hyperparameters whose empirical suitability must be evaluated separately.

Proposition C.6 (Moment Residuals of the Implemented Normalization). *Suppose both active token-sign groups are nonempty and all inputs are finite. Write*

$$P = \sum_{\mathcal{P}} A_{i,t}, \quad N = -\sum_{\mathcal{N}} A_{i,t}, \quad Q_+ = \sum_{\mathcal{P}} A_{i,t}^2, \quad Q_- = \sum_{\mathcal{N}} A_{i,t}^2,$$

and let $m = |\mathcal{P}| + |\mathcal{N}|$, $\kappa = P/N$, $D = Q_+ + \kappa^2 Q_-$, $\eta = \varepsilon_{\text{num}} > 0$. Before factor clamping, the implemented scales are

$$D_c = Q_+ + \min(\kappa^2 Q_-, C) + \eta, \quad \alpha_0 = \sqrt{m/D_c}, \quad \beta_0 = \alpha_0 \kappa,$$

where $C = 10^8$ is the intermediate cap. Their empirical mean is zero and their empirical second moment is D/D_c . In particular, if $\kappa^2 Q_- \leq C$, then the empirical variance v_0 satisfies

$$v_0 = \frac{D}{D + \eta} < 1, \quad 1 - v_0 = \frac{\eta}{D + \eta} \leq \frac{\eta}{D}. \quad (8)$$

In general,

$$\frac{D}{D_c} - 1 = \frac{\kappa^2 Q_- - \min(\kappa^2 Q_-, C) - \eta}{D_c},$$

so the product cap can change the second moment in either direction relative to one. Let a, b be the final clamped scales and let μ, v_2 be the resulting empirical mean and second moment on the m nonzero active tokens. Then

$$\begin{aligned} \mu &= \frac{aP - bN}{m}, & |\mu| &\leq \frac{|a - \alpha_0|P + |b - \beta_0|N}{m}, \\ v_2 &= \frac{a^2 Q_+ + b^2 Q_-}{m}, & \left| v_2 - \frac{D}{D_c} \right| &\leq \frac{|a - \alpha_0||a + \alpha_0|Q_+ + |b - \beta_0||b + \beta_0|Q_-}{m}. \end{aligned}$$

The empirical variance after factor clamping is $v_2 - \mu^2$. Positive clamping bounds preserve advantage signs and within-sign ordering. These statements describe the normalization branch in exact real arithmetic; when a fallback is taken, the active advantages are unchanged.

Proof. Partition the active index set into positive, negative, and zero advantages. The scaled first and second sums are $aP - bN$ and $a^2 Q_+ + b^2 Q_-$. Substituting $\beta_0 = \alpha_0 P/N$ gives zero first sum, and $\alpha_0^2 = m/D_c$ gives second moment D/D_c . The stabilizer deficit and product-cap identity follow by subtracting one. For the factor-clamp bounds, use

$$aP - bN = (a - \alpha_0)P - (b - \beta_0)N$$

and $a^2 - \alpha_0^2 = (a - \alpha_0)(a + \alpha_0)$, with the analogous identity for b , and apply the triangle inequality. Positivity of the clamped factors gives sign and within-sign order preservation. The finite-index moment identities, stabilizer and cap formulas, positive clamps, and residual bounds are verified in `formal/NormalizationSafeguards.lean`. $\square$

Fallback conditions. Normalization is skipped entirely when:

- One sign group is empty ($\mathcal{P} = \emptyset$ or $\mathcal{N} = \emptyset$). Assume, as in the token expansion convention above, that every sampled response contributes at least one active action token. Under the exact RLOO construction, the shaped rewards sum to zero within a prompt group, so this condition implies that every shaped reward is zero. The normalization is therefore skipped because its positive/negative constraints are infeasible; no gradient signal is created by this branch.
- The uniform-scale mode (Appendix C.11) is active and the implementation’s reward-uniformity tolerance is satisfied. In this branch the RLOO signal is intentionally replaced by the explicitly defined uniform-scale signal; the tolerance can include near-uniform rather than exactly equal rewards.

C.11 UNIFORM SCALE MODE

The uniform scale mode is an optional rule for using homogeneous reward groups. When the implementation’s reward-uniformity test is satisfied (exact equality is the ideal sparse-reward case; the configured tolerance can also include near-uniform rewards), the branch replaces the leave-one-out signal by the explicit rule below and skips adaptive normalization. For exactly identical rewards, the ordinary leave-one-out baseline would yield $\tilde{r}_i = 0$ for all i . The replacement gives nonzero coefficients only when the common reward is nonzero; an all-zero group remains zero. A nonzero coefficient alone does not establish a useful expected update. The population result below identifies a sufficient reward convention and its limitations. This statement is for the KL-free sparse-reward configuration: if an additional per-token KL or other shaping term is enabled, the post-shaping returns need not remain uniform and the branch must be analyzed with those augmented returns.

Definition C.7 (Uniform Scale). For prompt groups where $\text{std}(\{r_1, \dots, r_n\}) < \varepsilon_{\text{tol}}$, the implementation classifies the group as uniform-scale eligible, sets the shaped reward to $\tilde{r}_i = r_i/n$, and skips adaptive normalization. Exact equality of the rewards is a special case, not an additional premise of the scalar result.

Proposition C.8 (Signed coefficient in uniform-scale mode). *In the uniform-scale branch, a tolerance-selected reward group uses $A_i = r_i/n$ before the actual ProMax loss is evaluated. For an active token, write $a_t \geq 0$ for the action-mask/reduction weight, $M_t \in \{0, 1\}$ for the detached prefix gate, $w_t \geq 0$ for the detached rollout-to-snapshot weight, and $u_t = \exp(\log \pi_\theta - \log \pi_{\text{old}})$ for the differentiable PPO ratio. Away from the two PPO clipping kinks, the derivative of the scalar loss with respect to the current log-probability is*

$$-a_t M_t w_t u_t \frac{r_i}{n} I_t,$$

where $I_t = 1$ on the branch through which the current ratio differentiates and $I_t = 0$ when clipping or the detached gate removes that derivative. In particular, whenever $a_t, M_t, w_t, u_t > 0$ and the unclipped branch is active, the coefficient has the same sign as $-r_i$ (the sign for gradient descent therefore reinforces positive rewards and suppresses negative rewards). For $n > 0$, r_i/n is positive, zero, or negative exactly when r_i is positive, zero, or negative. This is a statement about each retained scalar score coefficient; it does not claim a coordinate-wise or full-vector gradient-direction guarantee.

Proof. With $\tilde{r}_i = r_i/n$, the implemented scalar token loss is the masked, importance-weighted PPO term, so its derivative must retain the prefix gate, detached IS coefficient, and clipping branch. The finite real-arithmetic derivative in the non-kink regions is checked by `ppoTerm_hasDerivAt` and `gated_ppo_hasFDerivAt` in `formal/PPODerivative.lean`; the finite sum over action positions is checked by `finite_gated_ppo_hasFDerivAt`. The simpler sign fact for the uniform coefficient is `uniform_scale_token_coefficient_sign`. Thus a positive reward reinforces the retained sampled score contribution and a negative reward suppresses it, subject to the gate and clipping branch. The exact effect on individual parameter coordinates depends on the score vectors. $\square$

Proposition C.9 (Population effect of the optional uniform-scale branch). *Fix a prompt and a differentiable finite response policy p_θ , with rewards in $\{r_-, r_+\}$, $r_- < r_+$, and $n \geq 2$ independent responses. Write $J_x(\theta) = \mathbb{E}_{p_\theta}[R]$, $s(\theta) = \Pr_{p_\theta}(R = r_+)$ and $u_\theta(y) = \nabla_\theta \log p_\theta(y)$, with the zero-mass score convention of the finite model. Assume the uniformity test selects exactly the homogeneous groups. At the on-policy point, with all prefix gates accepted and PPO clipping inactive, compare the sequence-score statistics*

$$g = \frac{1}{n} \sum_{i=1}^n \hat{A}_i u_\theta(y_i)$$

with uniform scale on and off. The mixed-group coefficients are identical in both cases and may include the actual asymmetric normalization and safeguards; coefficients are held fixed when computing the score update. Then

$$\mathbb{E}[g_{\text{on}} - g_{\text{off}}] = \lambda_n(s) \nabla_\theta s, \quad (9)$$

$$\lambda_n(s) := \frac{r_+ s^{n-1} - r_- (1-s)^{n-1}}{n}.$$

Since $\nabla_\theta J_x = (r_+ - r_-) \nabla_\theta s$, the additional expected term is aligned with the per-prompt reward gradient whenever $r_+ s^{n-1} \geq r_- (1-s)^{n-1}$. In particular, $r_- \leq 0 \leq r_+$ is sufficient. For 0/1 rewards, the extra term is $s^{n-1} \nabla_\theta J_x / n$.

Proof. RLOO is zero on a homogeneous group, including after its ordinary normalization fallback. Turning uniform scale on replaces that zero by r_+/n on the all-high event and r_-/n on the all-low event; other groups are unchanged. The independent response law gives

$$\mathbb{E} \left[\mathbb{I}_{\text{all high}} \sum_i u_\theta(y_i) \right] = n s^{n-1} \nabla_\theta s,$$

$$\mathbb{E} \left[\mathbb{I}_{\text{all low}} \sum_i u_\theta(y_i) \right] = -n (1-s)^{n-1} \nabla_\theta s.$$

These identities follow by separating each response from its $n - 1$ siblings in the finite product law. The success-restricted score expectation equals $\nabla_{\theta} s$ by finite policy differentiation; the failure-restricted expectation is its negative because the complete score has mean zero. Multiplying by the reward and the two factors $1/n$ proves Eq. equation 9. $\square$

This is a fixed-prompt, on-policy mechanism statement. The statistic sums each response’s token scores before any random total-token reduction. If every response has the same deterministic active length L , a global token mean multiplies both sides by $1/L$. A random denominator, off-policy weights, action-dependent masks, or active PPO clipping require the corrections in the general objective and reward certificate. Moreover, different prompts generally have different $\lambda_n(s)$, so the fixed-prompt result does not establish alignment after summing gradients of shared parameters across prompts. The result concerns the extra term; it does not assert that every finite stochastic optimizer step improves reward. When $0 < s < 1$, the probability of an all-success group is s^n , and all-failure 0/1 groups still produce zero coefficients.

Proposition C.10 (Complete binary update under class-dependent lengths). *Consider the same finite on-policy model with $n = m + 1 \geq 2$ iid responses, but now include the actual token reduction. Assume that every response with label $q \in \{0, 1\}$ has a deterministic active length L_q satisfying $0 < L_q \leq L_{\max}$. Let the positive/negative factors be computed from the length-weighted RLOO advantages using the implemented additive stabilizer, intermediate product cap, factor clamps and identity fallback, with numerical lower clamp $\eta \in (0, 1]$. Let H_i be the resulting coefficient of the response score after division by the sampled total active length. On-policy, with every prefix gate accepted and PPO clipping inactive, the complete expected sequence-score update satisfies*

$$\mathbb{E} \left[\sum_{i=1}^n H_i(Y_{1:n}) u_{\theta}(Y_i) \right] = \Lambda \nabla_{\theta} s, \quad \Lambda \geq \frac{\eta}{L_{\max}} > 0,$$

where $s = \Pr_{p_{\theta}}(R = 1)$. This holds both with uniform-scale disabled and with the optional 0/1 uniform-scale replacement on homogeneous groups. The multiplier may depend on p_{θ} , the sibling count and the lengths, but it is common to every parameter direction.

Proof. For a fixed coordinate i , condition on the m sibling labels. The implemented factors and the total token denominator are then fixed for the two possible labels of Y_i . The finite product law and the zero-mean score identity give the exact coordinate factorization

$$\mathbb{E}[H_i(Y_{1:n}) u_{\theta}(Y_i)] = \left(\sum_z p_{\theta}^{\otimes m}(z) [H_i(1, z) - H_i(0, z)] \right) \mathbb{E}[\mathbf{1}_{\{R=1\}} u_{\theta}(Y)].$$

For the binary RLOO signal, the sibling mean lies in $[0, 1]$. Positive factor clamps are at least η , while every sampled total length is at most $nL_{\max}$. Therefore each contrast is at least $\eta/(nL_{\max})$. The n coordinates give $\Lambda \geq \eta/L_{\max}$. On a homogeneous all-success group, the uniform replacement adds $1/(n \sum_i L_{Y_i})$ and has nonnegative contrast; on an all-failure 0/1 group it adds zero. Thus the lower bound is preserved. The finite conditioning, factorization, safeguard lower bound and the resulting common multiplier are checked by `coordinate_expectation`, `complete_update_expectation`, and `implementation_policy_direction` in `formal/BinaryUpdate.lean`. The label-dependent-length premise is explicit: if response length also depends on within-class content, the coefficient is no longer a function of the sibling labels alone and an additional covariance/error term is needed. $\square$

Proposition C.11 (Within-class length covariance decomposition). *Drop the class-dependent-length premise and let $C_i(y_{1:n})$ be the actual response-score coefficient produced by the implementation, including its length-weighted normalization, total-token denominator and optional uniform-scale term. For a fixed direction v , write $f_v(y) = \langle v, u_{\theta}(y) \rangle$. Assume both binary response classes have positive finite probability. Define Λ_C as the sum over coordinates and sibling draws of the difference between the conditional class means of C_i , and define $\mathcal{E}_v$ as the corresponding weighted within-class covariance between C_i and f_v . Then the exact identity is*

$$\mathbb{E} \left[\sum_i C_i(Y_{1:n}) f_v(Y_i) \right] = \Lambda_C \mathbb{E}[\mathbf{1}_{\{R=1\}} f_v(Y)] + \mathcal{E}_v,$$

and

$$|\mathcal{E}_v| \leq \sqrt{V_C n V_{f,v}},$$

Here, writing Z for the $n - 1$ independent sibling responses and inserting Y at coordinate i , define $\bar{C}_i(q, Z) := \mathbb{E}[C_i(Y, Z) \mid R(Y) = q, Z]$ and $\bar{f}_v(q) := \mathbb{E}[f_v(Y) \mid R(Y) = q]$. The variances are

$$V_C := \sum_i \mathbb{E}_{Z,Y}[(C_i(Y, Z) - \bar{C}_i(R(Y), Z))^2],$$

$$V_{f,v} := \mathbb{E}_Y[(f_v(Y) - \bar{f}_v(R(Y)))^2].$$

Therefore the complete update is guaranteed to have the same directional sign as the reward gradient whenever the positive main term dominates this explicit covariance margin. The bound does not assume independence between length, normalization and the response score.

Proof. Condition on the sibling responses and split the current response support into the two reward classes. Subtracting and adding the conditional class means of both C_i and f_v gives the displayed identity; the terms with one centered factor vanish within each class. Applying the finite Cauchy–Schwarz inequality to the remaining product gives the square-root bound. The exact decomposition is checked by `general_update_decomposition`, its finite covariance bound by `covariance_error_abs_bound`, and the implementation-specific coefficient is connected by `response_update_decomposition` in `formal/BinaryLength.lean`. If lengths are deterministic within each class, the coefficient is sibling-label measurable and $\mathcal{E}_v = 0$, recovering Proposition C.10. $\square$

Corollary C.12 (Local true-reward improvement of the complete binary update). *Fix a prompt, binary rewards $R \in \{0, 1\}$, and $n \geq 2$ iid responses from the finite differentiable policy above, with parameters in a real Hilbert space and a policy that is nonnegative and normalized in a neighborhood of θ . Assume $0 < s(\theta) < 1$, where $s(\theta) = \Pr_{p_\theta}(R = 1)$. At the on-policy point, let every prefix gate accept and PPO clipping be inactive. Allow arbitrary content-dependent active lengths $0 < L(y) \leq L_{\max}$ and retain the actual stabilizer, product cap, factor clamps and identity fallback. The lower factor clamp is $0 < \eta \leq 1$ and the upper clamp is at least η . The optional uniform-scale branch may be enabled or disabled; when enabled, assume its uniformity test selects exactly homogeneous groups. Using the actual response-score coefficients C_i and variances in Proposition C.11, set*

$$g := \nabla s(\theta), \quad U := \mathbb{E} \left[\sum_{i=1}^n C_i(Y_{1:n}) u_\theta(Y_i) \right], \quad f_g(y) := \langle g, u_\theta(y) \rangle.$$

The actual class multiplier satisfies $\Lambda_C \geq \eta/L_{\max} > 0$, and

$$\begin{aligned} \langle g, U \rangle &\geq \Lambda_C \|g\|^2 - \sqrt{V_C n V_{f,g}} \\ &\geq \frac{\eta}{L_{\max}} \|g\|^2 - \sqrt{V_C n V_{f,g}}. \end{aligned}$$

If $\sqrt{V_C n V_{f,g}} < \Lambda_C \|g\|^2$, there exists $r > 0$ such that every $0 < t < r$ satisfies

$$s(\theta + tU) > s(\theta).$$

The simpler condition $\sqrt{V_C n V_{f,g}} < \eta \|g\|^2 / L_{\max}$ is sufficient but can be more conservative than using the actual Λ_C . Under the class-dependent-length premise of Proposition C.10, $g \neq 0$ alone suffices for this local improvement: $U = \Lambda g$ with $\Lambda \geq \eta/L_{\max} > 0$.

Proof. For each fixed sibling draw, the sibling mean reward lies in $[0, 1]$. Positive normalization factors are bounded below by η , and the sampled total length is at most $nL_{\max}$. Averaging the resulting success/failure coefficient contrast within each class gives $\Lambda_C \geq \eta/L_{\max}$ even when length varies with content. The optional all-success replacement adds a non-negative contrast. Apply Proposition C.11 with $v = g$ and use the finite-policy identity $\mathbb{E}[\mathbf{1}_{\{R=1\}} f_g(Y)] = \|g\|^2$ to obtain the displayed lower bound. A positive right-hand side implies $\frac{d}{dt} s(\theta + tU) \Big|_{t=0} = \langle g, U \rangle > 0$. By differentiability, the difference quotient is positive for every sufficiently small positive t , proving strict improvement. For class-dependent lengths, Proposition C.10 directly gives $\langle g, U \rangle = \Lambda \|g\|^2 > 0$ whenever $g \neq 0$. The multiplier bound is checked by `response_class_multiplier_lower_bound` in `formal/`

BinaryLength.lean; the gradient connection, directional bound and both local improvement statements are checked by success_gradient_correct, population_ascent_bound, population_local_improvement, response_local_improvement and class_length_local_improvement in formal/BinaryAscent.lean. $\square$

This result concerns a sufficiently small step along the complete population update at a fixed prompt. It supplies no numerical step-size radius and does not imply improvement for every stochastic or Adam step. Arbitrary off-policy updates, active gates and clipping remain covered by the explicit correction terms in ??.

Reward translation and a complete counterexample. Adding a common offset c to both reward levels preserves the reward gradient and all raw RLOO signals, but changes the extra multiplier by

$$\lambda_n^{\text{shifted}}(s) - \lambda_n(s) = \frac{c}{n}(s^{n-1} - (1-s)^{n-1}).$$

For an explicit example, use two one-token responses per group, $p_\theta(\text{high}) = \theta$, $p_\theta(\text{low}) = 1 - \theta$, and rewards 4 and 3. At $\theta = 1/10$, $J(\theta) = 3 + \theta$, so $J'(\theta) = 1$. Mixed-group RLOO coefficients are $+1, -1$; their ideal normalization leaves them unchanged. The implemented real-arithmetic safeguards instead multiply both by

$$a_\varepsilon = \min \left\{ 10, \max \left\{ \varepsilon, \sqrt{\frac{2}{2+\varepsilon}} \right\} \right\}, \quad 0 < \varepsilon \leq 1,$$

so $0 < a_\varepsilon \leq 1$; the product cap is inactive. The mixed rewards have sample variance $1/2$, hence their standard deviation exceeds the implementation's 10^{-8} uniformity tolerance, while homogeneous groups have zero standard deviation. Thus this example follows the actual branch selection. Direct expectation under the two-response product law gives

$$\mathbb{E}[g_{\text{off}}] = a_\varepsilon > 0, \quad \mathbb{E}[g_{\text{on}}] = a_\varepsilon - \frac{23}{20} \leq -\frac{3}{20} < 0. \quad (10)$$

This is an expected ascent-direction reversal in a valid bounded-reward model, at an on-policy point with accepted gates and inactive clipping. It does not require a floating-point failure. With rewards 0, 1 in the same model, the corresponding expectation is $a_\varepsilon + \theta/2 > 0$ for $0 < \theta < 1$. The optional rule is therefore tied to the reward origin; these two reward conventions cannot be treated as equivalent ablations.

A constant reward carries no population signal. If every response in the policy support has reward c , uniform scale produces $g = (c/n^2) \sum_i u_\theta(y_i)$. The expected score is zero, so $\mathbb{E}[g] = 0$, as required by the constant reward objective. For every parameter direction v , independence gives

$$\text{Var}(\langle v, g \rangle) = \frac{c^2}{n^3} \mathbb{E}_p[\langle v, u_\theta(Y) \rangle^2].$$

Ordinary RLOO is identically zero in this setting. The optional branch can therefore introduce noise without useful population signal, while $c = 0$ leaves the entire update zero. The finite product-law expectation, differential connection, offset counterexample with stabilizer/clamp, and directional variance are checked in formal/UniformScale.lean.

D REINFORCE PRO: PROOFS AND VARIANTS

D.1 PROOF OF ??

We provide a complete proof of ?? under the definitions in ??. For convenience, write the log-threshold interval as

$$[\log(1 - \epsilon_{\text{low}}), \log(1 + \epsilon_{\text{high}})] = [-\tau_-, \tau_+], \quad \tau_+ := \log(1 + \epsilon_{\text{high}}), \quad \tau_- := -\log(1 - \epsilon_{\text{low}}),$$

so that $\rho_t \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ is equivalent to $\log \rho_t \in [-\tau_-, \tau_+]$.

Mask Definitions The three masks compared in ?? are:

- **Token-level (IcePop / token-level ratio mask):**

$$M_t^{\text{tok}} := \mathbb{I}[\rho_t \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]] = \mathbb{I}[\log \rho_t \in [-\tau_-, \tau_+]].$$

- **Sequence-level (GSPO / sequence-level ratio gate):**

$$M_t^{\text{seq}} := \mathbb{I}[\log \rho^{\text{seq}} \in [-\tau_-, \tau_+]], \quad \log \rho^{\text{seq}} := \frac{1}{T} \sum_{s=1}^T \log \rho_s,$$

for all t .

- **Prefix-level (Causal Trust Region proxy):**

$$M_t^{\text{pre}} := \mathbb{I}[\rho_t^{\text{pre}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]] = \mathbb{I}[\log \rho_t^{\text{pre}} \in [-\tau_-, \tau_+]],$$

where ρ_t^{pre} is defined in Eq. equation ??.

Proof of ??. **Part (a): Early deviation.** Assume $\rho_1 \notin [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ and $|\log \rho_t| \leq \varepsilon$ for all $t > 1$, with $0 \leq \varepsilon \leq \min(\tau_+, \tau_-)$. Then for every $t > 1$,

$$\log \rho_t \in [-\varepsilon, \varepsilon] \subseteq [-\tau_-, \tau_+],$$

so $M_t^{\text{tok}} = 1$ for all $t > 1$, while $M_1^{\text{tok}} = 0$ (since $\rho_1 \notin [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$). Hence token-level IcePop masks only position $t = 1$.

For the prefix mask, note first that $M_1^{\text{pre}} = 0$ because $\rho_1^{\text{pre}} = \rho_1 \notin [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$. To quantify how long the rejection can persist, use Eq. equation ??:

$$\log \rho_t^{\text{pre}} = \frac{1}{t} \sum_{s=1}^t \log \rho_s = \frac{1}{t} \left(\log \rho_1 + \sum_{s=2}^t \log \rho_s \right).$$

If $\log \rho_1 > \tau_+$, then $\log \rho_s \geq -\varepsilon$ for all $s \geq 2$, hence $\sum_{s=2}^t \log \rho_s \geq -(t-1)\varepsilon$ and therefore

$$\log \rho_t^{\text{pre}} \geq \frac{1}{t} (\log \rho_1 - (t-1)\varepsilon) = \frac{\log \rho_1 + \varepsilon}{t} - \varepsilon.$$

Thus, if $t < (\log \rho_1 + \varepsilon)/(\tau_+ + \varepsilon)$, then $\log \rho_t^{\text{pre}} > \tau_+$ and hence $M_t^{\text{pre}} = 0$. The lower-violation case $\log \rho_1 < -\tau_-$ is symmetric: since $\log \rho_s \leq \varepsilon$ for $s \geq 2$, we have $\sum_{s=2}^t \log \rho_s \leq \varepsilon(t-1)$ and therefore

$$\log \rho_t^{\text{pre}} \leq \frac{1}{t} (\log \rho_1 + \varepsilon(t-1)) = \frac{\log \rho_1 - \varepsilon}{t} + \varepsilon.$$

If $t < (-\log \rho_1 + \varepsilon)/(\tau_- + \varepsilon)$, then $\log \rho_t^{\text{pre}} < -\tau_-$ and hence $M_t^{\text{pre}} = 0$. This proves the sufficient conditions in ????

Part (b): Late deviation. Assume there exists $t^* \in \{1, \dots, T\}$ such that $\rho_{t^*} \notin [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$ and $|\log \rho_t| \leq \varepsilon$ for all $t \neq t^*$, with $0 \leq \varepsilon \leq \min(\tau_+, \tau_-)$. For any $t < t^*$, all active log-ratios in the prefix satisfy $\log \rho_s \in [-\tau_-, \tau_+]$. Since $\log \rho_t^{\text{pre}} = \frac{1}{t} \sum_{s=1}^t \log \rho_s$ is an average of values in $[-\tau_-, \tau_+]$, it follows that $\log \rho_t^{\text{pre}} \in [-\tau_-, \tau_+]$, and hence $M_t^{\text{pre}} = 1$ for all $t < t^*$. Therefore prefix masking can only occur for positions $t \geq t^*$, which implies

$$|\{t : M_t^{\text{pre}} = 0\}| \leq T - t^* + 1.$$

The sequence-level IS mask M_t^{seq} is constant across positions by definition, so it either rejects all tokens or rejects none.

For the claimed regime where prefix rejects but sequence-level accepts, consider the pure single-spike case where $\rho_t = 1$ for $t \neq t^*$ and $\rho_{t^*} > 1$. Then

$$\log \rho^{\text{seq}} = \frac{1}{T} \log \rho_{t^*}, \quad \log \rho_{t^*}^{\text{pre}} = \frac{1}{t^*} \log \rho_{t^*}.$$

Since $t^* \leq T$ and $\log \rho_{t^*} > 0$, we have $\log \rho_{t^*}^{\text{pre}} \geq \log \rho^{\text{seq}}$. Thus whenever

$$\frac{1}{t^*} \log \rho_{t^*} > \tau_+ \quad \text{and} \quad \frac{1}{T} \log \rho_{t^*} \leq \tau_+,$$

we obtain $M_{t^*}^{\text{pre}} = 0$ while $M_t^{\text{seq}} = 1$ for all t . For every $t^* < T$, the explicit choice $\log \rho_{t^*} = \tau_+(t^* + T)/2$ satisfies these inequalities, since

$$\tau_+ < \tau_+ \frac{t^* + T}{2t^*}, \quad 0 < \tau_+ \frac{t^* + T}{2T} < \tau_+.$$

For the lower threshold, choose $\log \rho_{t^*} = -\tau_-(t^* + T)/2$. Then

$$-\tau_- \frac{t^* + T}{2t^*} < -\tau_-, \quad -\tau_- < -\tau_- \frac{t^* + T}{2T} < 0,$$

which again rejects the spike under the prefix mask and accepts the whole sequence under the sequence mask. Both constructions have only one out-of-bound token. When $t^* = T$, the two statistics at that position are identical for every trajectory, so strict separation there is impossible.

Part (c): Sustained one-sided violation. If $\log \rho_s > \tau_+$ for all s , then for every t the prefix average satisfies $\log \rho_t^{\text{pre}} > \tau_+$, since it is an average of values each strictly larger than τ_+ . Each token mask also rejects directly by its defining interval test. Hence $M_t^{\text{tok}} = M_t^{\text{pre}} = 0$ for all t . The sequence-level average $\log \rho^{\text{seq}}$ is also strictly larger than τ_+ , so $M_t^{\text{seq}} = 0$ for all t . The lower-violation case $\log \rho_s < -\tau_-$ is symmetric. $\square$

D.2 COMPLETE FINITE-TREE ADAPTIVE BOUND

Proof of Theorem 10.10. Fix a prompt and write $p = \pi_{\text{roll}}$, $q = \pi_{\theta}$. For a history h with n actions remaining, let $V_n^p(h)$ and $V_n^q(h)$ be the respective expected terminal rewards. Let $S_n(h)$ be the rollout-occupancy surrogate for this continuation, and set $E_n(h) = V_n^q(h) - V_n^p(h) - S_n(h)$. Its local surrogate gain is $\sum_a (q(a | h) - p(a | h)) V_n^p(ha)$. Expanding the normalized policy recursions and collecting terms gives the exact identity

$$E_{n+1}(h) = \sum_a p(a | h) E_n(ha) + I_n(h),$$

$$I_n(h) := \sum_a (q(a | h) - p(a | h)) (V_n^q(ha) - V_n^p(ha)).$$

Here $E_0(h) = 0$. In particular, the recursion uses rollout probabilities in its outer expectation; no policy-occupancy substitution is assumed.

For either continuation starting at ha , bounded reward gives

$$|V_n^q(ha) - V_n^p(ha)| \leq 2R_{\max} D_{\text{TV}}(P^p(\cdot | ha), P^q(\cdot | ha)).$$

The continuation TV is at most 1 by normalization. Applying the expected-token-TV sum bound to this re-rooted tree gives $n\epsilon$. The KL chain rule on the same tree gives continuation reverse KL at most $n\delta$, so Pinsker gives the third bound $\sqrt{n\delta/2}$. All required local contexts have length less than T ; there is no assumption on divergences beyond the generation horizon. Define $b_n = \min(1, n\epsilon, \sqrt{n\delta/2})$. Then

$$|I_n(h)| \leq 2R_{\max} b_n \sum_a |q(a | h) - p(a | h)| = 4R_{\max} b_n D_{\text{TV}}^{\text{tok}}(h; p, q).$$

Unrolling the error recursion from $E_T(\emptyset)$ yields a sum of these local interactions, averaged over rollout histories. At position t , the continuation has $T - t$ actions. Taking absolute values and then averaging over the finite prompt distribution therefore gives

$$|\text{Error}(\pi_{\theta})| \leq 4R_{\max} \sum_{t=1}^T \bar{D}_t b_{T-t}.$$

The performance lower bound follows by subtracting this error allowance from the surrogate. The trajectory identity in Remark E.8 identifies E_T with the negative reward-weighted remainder, so the same absolute bound applies to that remainder. Shared absorbing EOS transitions have zero local TV and zero interaction. $\square$

The exact recursion is checked by `surrogate_error_future_identity`. The three future-TV bounds are combined in `futureTV_le_adaptive`, and `surrogate_error_adaptive_root` and `trajectory_remainder_adaptive` prove the root error and actual trajectory remainder bounds. `family_remainder_adaptive` and `family_performance_lower_bound_adaptive` include the finite prompt law and the paper’s expected per-position TV, in `formal/AdaptiveBound.lean`.

D.3 PREFIX CAUSAL TRUST REGION MECHANISM BOUNDS

This subsection proves `?????`, which formalize a per-position mechanism interpretation of prefix Causal Trust Region masking.

Proof of ??. Fix $t \in \{1, \dots, T\}$ and a reachable context $c_{t+1} = (x, y_{\leq t})$ with $d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) > 0$. By the definition of the context visitation distribution, for any policy π ,

$$\begin{aligned} d_{t+1}^{\pi}(c_{t+1}) &= P(x) P^{\pi}(y_{\leq t} \mid x) \\ &= P(x) \prod_{s=1}^t \pi(y_s \mid c_s), \end{aligned}$$

where $c_s = (x, y_{<s})$ and the second line is the autoregressive factorization. Under the mutual-support Assumption 10.3, the ratio $d_{t+1}^{\pi_{\theta}}(c_{t+1})/d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})$ is well-defined for all such contexts. Therefore,

$$\begin{aligned} \frac{d_{t+1}^{\pi_{\theta}}(c_{t+1})}{d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})} &= \prod_{s=1}^t \frac{\pi_{\theta}(y_s \mid c_s)}{\pi_{\text{roll}}(y_s \mid c_s)} \\ &= \prod_{s=1}^t \rho_s = W_t, \end{aligned}$$

as claimed. $\square$

Proof of ??. Fix $t \in \{1, \dots, T\}$. By definition of total variation distance,

$$\|d_{t+1}^{\pi_{\theta}} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} = \frac{1}{2} \sum_{c_{t+1}} |d_{t+1}^{\pi_{\theta}}(c_{t+1}) - d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})|.$$

By `??`, for every reachable c_{t+1} with $d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) > 0$,

$$d_{t+1}^{\pi_{\theta}}(c_{t+1}) = d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) W_t.$$

Substituting this into the previous display yields

$$\begin{aligned} \|d_{t+1}^{\pi_{\theta}} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} &= \frac{1}{2} \sum_{c_{t+1}} d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) |W_t - 1| \\ &= \frac{1}{2} \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [|W_t - 1|], \end{aligned}$$

which proves the claim. $\square$

Proof of ??. Fix $t \in \{1, \dots, T\}$. Recall that $\tau_+ := \log(1 + \epsilon_{\text{high}})$ and $\tau_- := -\log(1 - \epsilon_{\text{low}})$.

Step 1: Rewrite the acceptance event. By definition of the prefix mask in Eq. equation `??`,

$$M_t^{\text{pre}} = 1 \iff \log \rho_t^{\text{pre}} \in [-\tau_-, \tau_+].$$

Using Eq. equation `??`, this is equivalent to the cumulative log-ratio constraint

$$-t\tau_- \leq \sum_{s=1}^t \log \rho_s \leq t\tau_+.$$

Step 2: Convert to a bound on the prefix density ratio. Recall that $W_t = \prod_{s=1}^t \rho_s$. Equivalently,

$$W_t = \exp\left(\sum_{s=1}^t \log \rho_s\right).$$

Therefore, the previous display implies that, on $\{M_t^{\text{pre}} = 1\}$,

$$e^{-t\tau_-} \leq W_t \leq e^{t\tau_+}.$$

Step 3: Bound the ratio residual and take expectations. If a scalar w lies in an interval $[a, b]$ with $0 < a \leq b$, then $|w - 1| \leq \max\{b - 1, 1 - a\}$. Applying this with $a = e^{-t\tau_-}$ and $b = e^{t\tau_+}$ yields that on $\{M_t^{\text{pre}} = 1\}$,

$$|W_t - 1| \leq \max\{e^{t\tau_+} - 1, 1 - e^{-t\tau_-}\}.$$

Multiplying by M_t^{pre} and taking expectations under $c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}$ gives

$$\begin{aligned} \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [M_t^{\text{pre}} |W_t - 1|] &\leq \max\{e^{t\tau_+} - 1, 1 - e^{-t\tau_-}\} \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [M_t^{\text{pre}}] \\ &\leq \max\{e^{t\tau_+} - 1, 1 - e^{-t\tau_-}\}, \end{aligned}$$

which completes the proof. $\square$

The complete finite-history statement is checked in `formal/TreePrefixGate.lean`. The gate is defined by the actual cumulative log-ratio divided by the active prefix length. `accepted_prefix_density_envelope` proves the density-ratio and residual bounds on positive-mass accepted prefixes; `prefix_gate_residual_bound` and `prefix_gate_residual_le_envelope` prove both expectation inequalities under the normalized rollout tree. `prefix_gate_tv_bound` also identifies the corresponding bound on the half-L1 distance of the filtered prefix masses. These statements allow an action-dependent gate and common zero-mass tokens. They concern filtered probability mass and do not assert an unbiased masked policy gradient.

D.4 SUB-GAUSSIAN STRENGTHENING: PER-POSITION BOUNDS VIA PREFIX DENSITY RATIOS

This subsection proves Lemmas D.2 and D.6, Corollaries D.3, D.7 and D.8, and Theorem D.4, which strengthen the prefix density-ratio mechanism view with explicit per-position bounds under a conditional concentration assumption.

D.4.1 STATEMENTS AND INTERPRETATION

The mechanism in ?? is distributional: it identifies W_t as the change-of-measure object that controls context shift. To turn ?? into an explicit per-position bound, we introduce a quantitative concentration assumption on the per-token log-ratio increments.

Assumption D.1 (Conditionally sub-Gaussian log-ratio increments). Under the rollout distribution $y_t \sim \pi_{\text{roll}}(\cdot | c_t)$, define the centered increment

$$X_t := \log \rho_t - \mathbb{E}[\log \rho_t | c_t].$$

Assume that $\log \rho_t$ is conditionally integrable and that $D_{\text{KL}}(\pi_{\text{roll}}(\cdot | c_t) \| \pi_{\theta}(\cdot | c_t)) < \infty$ on every reachable context. There exists a constant $\sigma^2 > 0$ such that for all $t \in \{1, \dots, T\}$, all reachable contexts c_t , and all $\lambda \in \mathbb{R}$,

$$\mathbb{E}[\exp(\lambda X_t) | c_t] \leq \exp\left(\frac{\lambda^2 \sigma^2}{2}\right).$$

Lemma D.2 (Second-moment control of the prefix density ratio). *In the finite-vocabulary history-tree model, assume the mutual-support Assumption 10.3 and Assumption D.1. Then for any $t \in \{1, \dots, T\}$,*

$$\mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [W_t^2] \leq \exp(2\sigma^2 t).$$

Corollary D.3 (An explicit per-position context-shift bound). *In the same finite-vocabulary history-tree model, assume the mutual-support Assumption 10.3 and Assumption D.1. Then for any $t \in \{1, \dots, T\}$,*

$$\|d_{t+1}^{\pi_{\theta}} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\exp(2\sigma^2 t) - 1}.$$

Theorem D.4 (A per-position surrogate-error bound under sub-Gaussian prefix drift). *In the same finite-vocabulary history-tree model, assume the mutual-support Assumption 10.3, Assumption D.1, and bounded rewards (i.e., $|R(x, y)| \leq R_{\max}$). Then the surrogate error in Eq. equation 5 satisfies*

$$|\text{Error}(\pi_\theta)| \leq 2R_{\max} \sum_{t=1}^T \sqrt{\exp(2\sigma^2(t-1)) - 1}.$$

On the finite history tree, the proof expands the cumulative advantage signal into its expectations under each prefix distribution, applies the concentration bound to their differences, and uses $|g_t| \leq 2R_{\max}$. This complete finite-model argument is verified by `surrogate_error_of_mgf` in `formal/PrefixConcentration.lean`. General measurable prompt spaces, almost-sure conditional assumptions, and the general filtration formulation remain separate formalization obligations.

Corollary D.5 (A concentration-based improvement certificate). *For each fixed prompt in the finite-vocabulary model, under the assumptions of Theorem D.4, define*

$$B_T(\sigma) := 2R_{\max} \sum_{t=1}^T \sqrt{\exp(2\sigma^2(t-1)) - 1}.$$

For this fixed prompt x , write $J_x(\pi) := \mathbb{E}_{y \sim P^\pi(\cdot|x)}[R(x, y)]$ and let $\mathcal{L}_{x, \pi_{\text{roll}}}^{\text{LLM}}(\pi_\theta)$ denote the corresponding rollout surrogate with the prompt held fixed. Then

$$J_x(\pi_\theta) - J_x(\pi_{\text{roll}}) \geq \mathcal{L}_{x, \pi_{\text{roll}}}^{\text{LLM}}(\pi_\theta) - B_T(\sigma).$$

In particular, $\mathcal{L}_{x, \pi_{\text{roll}}}^{\text{LLM}}(\pi_\theta) > B_T(\sigma)$ implies strict improvement in expected reward.

Proof. Apply the lower side of the absolute surrogate-error bound in Theorem D.4. The lower bound and strict-improvement implication are checked by `performance_lower_bound_of_mgf` and `performance_improves_of_mgf` in `formal/PrefixConcentration.lean`. $\square$

The certificate concerns the raw population surrogate. ?? connects the actual ProMax objective to that surrogate using explicit gate, normalization, clipping and sampling terms. Its estimation argument also accepts the concentration error bound above in place of the Adaptive cost whenever those assumptions hold.

Acceptance-coupled filtered drift bounds. The Pro Max loss (Eq. equation ??) multiplies each per-token surrogate term by the prefix gate M_t^{pre} , so gradient signal is received only from *accepted* prefixes. To quantify policy drift on this filtered subset, define the accepted-prefix context measure (a sub-probability measure)

$$\bar{d}_{t+1}^\pi(c_{t+1}) := d_{t+1}^\pi(c_{t+1}) M_t^{\text{pre}}(c_{t+1}), \quad t \in \{1, \dots, T\},$$

where we view M_t^{pre} as a deterministic function of the length- t prefix $c_{t+1} = (x, y_{\leq t})$ under the fixed policy pair $(\pi_{\text{roll}}, \pi_\theta)$. For these possibly unequal-mass measures, we explicitly use $\|\mu - \nu\|_{\text{TV}} := \frac{1}{2} \sum_c |\mu(c) - \nu(c)|$, the half- L^1 convention; this need not equal $\sup_B |\mu(B) - \nu(B)|$ when the total masses differ. No renormalization is applied. Its total mass under rollout is the acceptance rate

$$\alpha_t := \mathbb{E}_{c_{t+1} \sim \bar{d}_{t+1}^{\pi_{\text{roll}}}}[M_t^{\text{pre}}] = \sum_{c_{t+1}} \bar{d}_{t+1}^{\pi_{\text{roll}}}(c_{t+1}).$$

Lemma D.6 (TV identity for accepted-prefix context measures). *In the finite-vocabulary, normalized history-tree model used above, assume the mutual-support Assumption 10.3. Then for any $t \in \{1, \dots, T\}$,*

$$\|\bar{d}_{t+1}^{\pi_\theta} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} = \frac{1}{2} \mathbb{E}_{c_{t+1} \sim \bar{d}_{t+1}^{\pi_{\text{roll}}}}[M_t^{\text{pre}} |W_t - 1|].$$

Corollary D.7 (Filtered drift bound with acceptance rate). *In the same finite-vocabulary, normalized history-tree model, assume the mutual-support Assumption 10.3 and Assumption D.1. Then for any $t \in \{1, \dots, T\}$,*

$$\mathbb{E}_{c_{t+1} \sim \bar{d}_{t+1}^{\pi_{\text{roll}}}}[M_t^{\text{pre}} |W_t - 1|] \leq \sqrt{\alpha_t} \sqrt{\exp(2\sigma^2 t) - 1}.$$

Equivalently,

$$\|\bar{d}_{t+1}^{\pi_\theta} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\alpha_t} \sqrt{\exp(2\sigma^2 t) - 1}.$$

Corollary D.8 (Gate-induced effective horizon). *In the same finite-vocabulary, normalized history-tree model, assume the mutual-support Assumption 10.3 and Assumption D.1. Define the effective horizon*

$$H_{\text{eff}} := \sum_{t=1}^T \alpha_t = \sum_{t=1}^T \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [M_t^{\text{pre}}].$$

Then the cumulative accepted-prefix drift satisfies

$$\begin{aligned} \sum_{t=1}^T \|\bar{d}_{t+1}^{\pi_{\theta}} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} &\leq \frac{1}{2} \sum_{t=1}^T \sqrt{\alpha_t} \sqrt{\exp(2\sigma^2 t) - 1} \\ &\leq \frac{1}{2} \sqrt{H_{\text{eff}}} \sqrt{\sum_{t=1}^T (\exp(2\sigma^2 t) - 1)}. \end{aligned}$$

In particular, Corollary D.8 makes explicit the $\sqrt{\alpha_t}$ per-position attenuation and summarizes its aggregate effect through H_{eff} , formalizing a “gate + concentration” effective-horizon view. The attenuation includes the effect of reduced accepted mass; it does not assert that the renormalized conditional distributions become closer.

These accepted-prefix quantities are a filtered drift diagnostic for the policy pair that supplies the prefix ratio (the implementation pair is $(\pi_{\text{roll}}, \pi_{\text{old}})$). They coincide with the behavior/current-policy pair at the start of an update when $\pi_{\text{old}} = \pi_{\theta}$. Because M_t^{pre} depends on the sampled prefix including y_t , it is measurable with respect to c_{t+1} rather than c_t ; therefore the displayed TV identities do not by themselves assert an unbiased policy-gradient identity for the masked objective. The mask is a causal sample-level filter, and a direct bound on the masked gradient would require an additional change-of-measure theorem on the action-extended prefix space.

The remaining proofs follow below.

For a finite vocabulary and horizon, the concentration argument is verified on the full autoregressive history tree in `formal/PrefixConcentration.lean`. The construction multiplies normalized, history-dependent token kernels to obtain prefix masses, proves their normalization and density-ratio identity, and connects the resulting finite expectations to the recursive second moment. Common zero-probability actions, including shared absorbing EOS transitions, are allowed under mutual support. No independence across tokens is assumed. The centered-MGF condition is needed only at $\lambda = 2$ and only at rollout-reachable histories shorter than the chosen horizon; zero-probability branches vanish in the second-moment recursion. The all- λ assumption below implies this condition. The conditional mean is proved non-positive from the local policy probabilities, rather than supplied as an additional hypothesis. The same development proves the prefix and accepted-prefix TV bounds, their effective-horizon aggregation, and the resulting surrogate-error bound in Lemmas D.2 and D.6, Corollaries D.3, D.7 and D.8, and Theorem D.4. The exact theorem names and remaining general probability-space interfaces are recorded in `formal/PROOF_AUDIT.md`.

D.4.2 PROOFS

Proof of Lemma D.2. Fix $t \in \{1, \dots, T\}$. Throughout this proof, expectations are taken under the rollout distribution $c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}$ (equivalently, $(x, y_{\leq t})$ generated by π_{roll}). Recall $W_t = \prod_{s=1}^t \rho_s$.

Step 1: Decompose log-ratios into mean + centered increments. For each $s \in \{1, \dots, t\}$, define

$$\mu_s := \mathbb{E}[\log \rho_s \mid c_s], \quad X_s := \log \rho_s - \mu_s.$$

Under mutual support and the conditional integrability in Assumption D.1, the conditional expectation is well-defined, and moreover

$$\mu_s = \mathbb{E}_{y_s \sim \pi_{\text{roll}}(\cdot \mid c_s)} \left[\log \frac{\pi_{\theta}(y_s \mid c_s)}{\pi_{\text{roll}}(y_s \mid c_s)} \right] = -D_{\text{KL}}(\pi_{\text{roll}}(\cdot \mid c_s) \parallel \pi_{\theta}(\cdot \mid c_s)) \leq 0.$$

Therefore,

$$\begin{aligned} W_t^2 &= \exp\left(2 \sum_{s=1}^t \log \rho_s\right) \\ &= \exp\left(2 \sum_{s=1}^t \mu_s\right) \cdot \exp\left(2 \sum_{s=1}^t X_s\right) \\ &\leq \exp\left(2 \sum_{s=1}^t X_s\right), \end{aligned}$$

and hence

$$\mathbb{E}[W_t^2] \leq \mathbb{E}\left[\exp\left(2 \sum_{s=1}^t X_s\right)\right].$$

Step 2: Control the MGF of the adapted sum by tower property. Let $\mathcal{F}_s := \sigma(x, y_{\leq s})$ denote the natural filtration. Since $c_s = (x, y_{\leq s})$ is $\mathcal{F}_{s-1}$ -measurable and X_s depends only on (c_s, y_s) , we have

$$\mathbb{E}[\exp(2X_s) \mid \mathcal{F}_{s-1}] = \mathbb{E}[\exp(2X_s) \mid c_s].$$

By Assumption D.1 with $\lambda = 2$,

$$\mathbb{E}[\exp(2X_s) \mid c_s] \leq \exp\left(\frac{2^2 \sigma^2}{2}\right) = \exp(2\sigma^2).$$

Therefore, applying the tower property yields

$$\begin{aligned} \mathbb{E}\left[\exp\left(2 \sum_{s=1}^t X_s\right)\right] &= \mathbb{E}\left[\exp\left(2 \sum_{s=1}^{t-1} X_s\right) \cdot \mathbb{E}[\exp(2X_t) \mid \mathcal{F}_{t-1}]\right] \\ &\leq \exp(2\sigma^2) \mathbb{E}\left[\exp\left(2 \sum_{s=1}^{t-1} X_s\right)\right]. \end{aligned}$$

Iterating this inequality gives

$$\mathbb{E}\left[\exp\left(2 \sum_{s=1}^t X_s\right)\right] \leq \exp(2\sigma^2 t).$$

Combining with Step 1 completes the proof. $\square$

Proof of Corollary D.3. Fix $t \in \{1, \dots, T\}$. By ??,

$$\|d_{t+1}^{\pi_\theta} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} = \frac{1}{2} \mathbb{E}[|W_t - 1|].$$

By Cauchy–Schwarz, $\mathbb{E}[|Z|] \leq \sqrt{\mathbb{E}[Z^2]}$, so

$$\mathbb{E}[|W_t - 1|] \leq \sqrt{\mathbb{E}[(W_t - 1)^2]}.$$

Expanding the square yields

$$\mathbb{E}[(W_t - 1)^2] = \mathbb{E}[W_t^2] - 2\mathbb{E}[W_t] + 1.$$

By ??, $W_t = d_{t+1}^{\pi_\theta}(c_{t+1})/d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})$, so

$$\mathbb{E}[W_t] = \sum_{c_{t+1}} d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) \frac{d_{t+1}^{\pi_\theta}(c_{t+1})}{d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})} = \sum_{c_{t+1}} d_{t+1}^{\pi_\theta}(c_{t+1}) = 1.$$

Together with Lemma D.2, this implies

$$\mathbb{E}[(W_t - 1)^2] \leq \exp(2\sigma^2 t) - 1.$$

Substituting into the first display gives the claim. $\square$

Proof of Theorem D.4. Recall the error decomposition in Eq. equation 5:

$$|\text{Error}(\pi_\theta)| \leq \sum_{t=1}^T 2\|g_t\|_\infty \cdot \|d_t^{\pi_\theta} - d_t^{\pi_{\text{roll}}}\|_{\text{TV}}.$$

Step 1: Bound the advantage factor by bounded rewards. Fix t and a context c_t . By definition,

$$g_t(c_t) := \mathbb{E}_{y_t \sim \pi_\theta(\cdot | c_t)}[A_t^{\pi_{\text{roll}}}(c_t, y_t)],$$

$$A_t^{\pi_{\text{roll}}}(c_t, y_t) := \mathbb{E}_{\pi_{\text{roll}}}[R(x, y) | c_t, y_t] - \mathbb{E}_{\pi_{\text{roll}}}[R(x, y) | c_t].$$

Under bounded rewards $|R(x, y)| \leq R_{\max}$, both conditional expectations lie in $[-R_{\max}, R_{\max}]$, hence $|A_t^{\pi_{\text{roll}}}(c_t, y_t)| \leq 2R_{\max}$ and consequently $|g_t(c_t)| \leq 2R_{\max}$. Therefore $\|g_t\|_\infty \leq 2R_{\max}$.

Step 2: Apply the per-position TV bound. For each $t \in \{1, \dots, T\}$, observe that d_t is the context distribution indexed by token position t , i.e., it corresponds to a length- $(t-1)$ prefix. Thus Corollary D.3 with index $(t-1)$ gives

$$\|d_t^{\pi_\theta} - d_t^{\pi_{\text{roll}}}\|_{\text{TV}} \leq \frac{1}{2} \sqrt{\exp(2\sigma^2(t-1)) - 1}.$$

Step 3: Combine. Substituting the bounds from Steps 1–2 into the error decomposition yields

$$|\text{Error}(\pi_\theta)| \leq \sum_{t=1}^T 2(2R_{\max}) \cdot \frac{1}{2} \sqrt{\exp(2\sigma^2(t-1)) - 1}$$

$$= 2R_{\max} \sum_{t=1}^T \sqrt{\exp(2\sigma^2(t-1)) - 1},$$

which completes the proof. $\square$

Proof of Lemma D.6. Fix $t \in \{1, \dots, T\}$. Recall the accepted-prefix context measure

$$\bar{d}_{t+1}^\pi(c_{t+1}) := d_{t+1}^\pi(c_{t+1}) M_t^{\text{pre}}(c_{t+1}).$$

By definition of total variation distance,

$$\|\bar{d}_{t+1}^{\pi_\theta} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} = \frac{1}{2} \sum_{c_{t+1}} |\bar{d}_{t+1}^{\pi_\theta}(c_{t+1}) - \bar{d}_{t+1}^{\pi_{\text{roll}}}(c_{t+1})|$$

$$= \frac{1}{2} \sum_{c_{t+1}} M_t^{\text{pre}}(c_{t+1}) |d_{t+1}^{\pi_\theta}(c_{t+1}) - d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})|.$$

By ??, for every reachable c_{t+1} with $d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) > 0$ one has $d_{t+1}^{\pi_\theta}(c_{t+1}) = d_{t+1}^{\pi_{\text{roll}}}(c_{t+1})W_t$. Substituting yields

$$\|\bar{d}_{t+1}^{\pi_\theta} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} = \frac{1}{2} \sum_{c_{t+1}} d_{t+1}^{\pi_{\text{roll}}}(c_{t+1}) M_t^{\text{pre}}(c_{t+1}) |W_t - 1|$$

$$= \frac{1}{2} \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [M_t^{\text{pre}} |W_t - 1|],$$

which proves the claim. $\square$

Proof of Corollary D.7. Fix $t \in \{1, \dots, T\}$ and write $\alpha_t := \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [M_t^{\text{pre}}]$. By Cauchy-Schwarz,

$$\mathbb{E}[M_t^{\text{pre}} |W_t - 1|] \leq \sqrt{\mathbb{E}[M_t^{\text{pre}}]} \sqrt{\mathbb{E}[(W_t - 1)^2]} = \sqrt{\alpha_t} \sqrt{\mathbb{E}[(W_t - 1)^2]}.$$

As in the proof of Corollary D.3, one has $\mathbb{E}[W_t] = 1$ and hence

$$\mathbb{E}[(W_t - 1)^2] = \mathbb{E}[W_t^2] - 1.$$

By Lemma D.2, $\mathbb{E}[W_t^2] \leq \exp(2\sigma^2 t)$, so $\mathbb{E}[(W_t - 1)^2] \leq \exp(2\sigma^2 t) - 1$. Substituting into the previous display gives

$$\mathbb{E}[M_t^{\text{pre}} |W_t - 1|] \leq \sqrt{\alpha_t} \sqrt{\exp(2\sigma^2 t) - 1}.$$

The equivalent TV bound follows immediately from Lemma D.6. $\square$

Proof of Corollary D.8. Define $\alpha_t := \mathbb{E}_{c_{t+1} \sim d_{t+1}^{\pi_{\text{roll}}}} [M_t^{\text{pre}}]$ and $H_{\text{eff}} := \sum_{t=1}^T \alpha_t$. Applying Corollary D.7 for each t and summing yields

$$\sum_{t=1}^T \|\bar{d}_{t+1}^{\pi_{\theta}} - \bar{d}_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} \leq \frac{1}{2} \sum_{t=1}^T \sqrt{\alpha_t} \sqrt{\exp(2\sigma^2 t) - 1}.$$

For the compressed effective-horizon form, apply Cauchy–Schwarz to the right-hand side with $b_t := \sqrt{\exp(2\sigma^2 t) - 1}$:

$$\begin{aligned} \sum_{t=1}^T \sqrt{\alpha_t} b_t &\leq \sqrt{\sum_{t=1}^T \alpha_t} \sqrt{\sum_{t=1}^T b_t^2} \\ &= \sqrt{H_{\text{eff}}} \sqrt{\sum_{t=1}^T (\exp(2\sigma^2 t) - 1)}, \end{aligned}$$

which proves the claim. $\square$

D.5 ALTERNATIVE TRUST-REGION MASKING VARIANTS

This appendix collects several masking variants that are compatible with the ratio setup in the main text (??). Throughout, samples are generated by the behavior policy π_{roll} and gradients are computed under the training policy π_{θ} . We write the per-token importance ratio as

$$\rho_t := \frac{\pi_{\theta}(y_t | c_t)}{\pi_{\text{roll}}(y_t | c_t)}.$$

We also reuse the prefix statistic ρ_t^{pre} from ??. For thresholds $\epsilon_{\text{low}} \in (0, 1)$ and $\epsilon_{\text{high}} > 0$, define the log-threshold interval

$$[\log(1 - \epsilon_{\text{low}}), \log(1 + \epsilon_{\text{high}})] = [-\tau_-, \tau_+], \quad \tau_+ := \log(1 + \epsilon_{\text{high}}), \quad \tau_- := -\log(1 - \epsilon_{\text{low}}).$$

Variant I: Absorbing Prefix-Average Mask The main text uses a pointwise prefix gate at each position t . A more rejection-style variant enforces an *absorbing* semantics: once a prefix violates the threshold, all subsequent positions are rejected.

$$\widetilde{M}_t^{\text{pre}} := \mathbb{I}[\rho_t^{\text{pre}} \in [1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]], \quad M_t^{\text{abs-pre}} := \prod_{u=1}^t \widetilde{M}_u^{\text{pre}}.$$

Variant II: Prefix-Max Mask / Max-Filtering Max-based trust-region criteria can be expressed via a sample-based proxy $k_2(\rho)$:

$$k_2(\rho) := \frac{1}{2} (\log \rho)^2.$$

The corresponding prefix-max statistic and mask are

$$S_t^{\text{max}} := \max_{1 \leq s \leq t} k_2(\rho_s), \quad M_t^{\text{pre-max}} := \mathbb{I}[S_t^{\text{max}} \leq \delta_{\text{max}}].$$

Equivalently, $M_t^{\text{pre-max}} = 1$ iff $\max_{s \leq t} |\log \rho_s| \leq \sqrt{2\delta_{\text{max}}}$.

Variant III: Prefix-Score Mask / Bound-Driven Score-based masking controls a prefix-level aggregate of per-token ratio deviations. Define a nonnegative per-token deviation

$$a_t := |\rho_t - 1|$$

and deterministic nonnegative weights $(w_t)_{t \geq 1}$ with $\sum_{s \leq t} w_s > 0$. The prefix score is

$$\widetilde{W}_t := \frac{\sum_{s=1}^t w_s a_s}{\sum_{s=1}^t w_s},$$

and the corresponding mask is

$$M_t^{\text{pre-score}} := \mathbb{I}[\widetilde{W}_t \leq \delta_W].$$

A special case is the uniform-weight choice $w_s \equiv 1$, which yields $\widetilde{W}_T = \frac{1}{T} \sum_{t=1}^T |\rho_t - 1|$ at the full horizon.

Variant IV: Prefix Abnormal-Token Rate / TRM-Style TRM (Li et al., 2025) uses per-token divergence estimates (or their sample-based proxies) to decide whether a trajectory lies inside a trust region. We can adapt this idea to a prefix-level gate by counting *abnormal* tokens in the prefix. Concretely, define the abnormal-token indicator

$$B_t := \mathbb{I}[D_{\text{KL}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta}) > \delta_{\text{TRM}}],$$

and the prefix abnormal count and rate

$$N_t := \sum_{s=1}^t B_s, \quad \hat{p}_t := \frac{N_t}{t}.$$

Given a rate threshold $\xi \in [0, 1]$, the corresponding prefix mask is

$$M_t^{\text{pre-anom}} := \mathbb{I}[\hat{p}_t \leq \xi].$$

In practice, one may instantiate B_t using TRM’s computation method for $D_{\text{KL}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta})$; for example, via the sample-based proxy $k_3(\rho) := \rho - 1 - \log \rho$. Under the mutual-support Assumption 10.3, the ratio definition $\rho_t = \pi_{\theta}/\pi_{\text{roll}}$ gives

$$\mathbb{E}_{\pi_{\text{roll}}}[k_3(\rho_t) \mid c_t] = D_{\text{KL}}(\pi_{\text{roll}}(\cdot \mid c_t) \parallel \pi_{\theta}(\cdot \mid c_t)).$$

The finite-support normalization algebra behind this equality is checked by the Lean theorem `finite_k3_proxy_eq_reverse_kl`; its general conditional-expectation version still requires the probability-space formalization listed in the audit report. The following inverse certificate connects a sample-level k_3 threshold to the inlier condition used below. A threshold on the conditional mean KL does not give this pointwise implication for each sampled token.

Lemma D.9 (A Log-Ratio Interval from the Sample k_3 Proxy). *Write $F(z) = e^z - 1 - z$, so $k_3(\rho) = F(\log \rho)$ for $\rho > 0$. For $\tau_-, \tau_+ \geq 0$, if*

$$k_3(\rho) \leq \delta \quad \text{and} \quad \delta \leq \min\{F(-\tau_-), F(\tau_+)\},$$

then $-\tau_- \leq \log \rho \leq \tau_+$. In particular, for any $\tau_0 > 0$, the threshold

$$\delta_0(\tau_0) := \min\{e^{-\tau_0} - 1 + \tau_0, e^{\tau_0} - 1 - \tau_0\} > 0$$

ensures that $B_s = \mathbb{I}[k_3(\rho_s) > \delta]$ with $0 \leq \delta \leq \delta_0(\tau_0)$ satisfies $B_s = 0 \Rightarrow |\log \rho_s| \leq \tau_0$.

Proof. F is strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0, \infty)$, with $F(0) = 0$. If $\log \rho < -\tau_-$ or $\log \rho > \tau_+$, then $F(\log \rho) > \delta$, a contradiction. The same strict monotonicity gives positivity of both terms defining $\delta_0(\tau_0)$. The interval certificate, strict monotonicity, and positive threshold are checked in `formal/K3Inlier.lean`. $\square$

Lemma D.10 (From Abnormal-Token Rate to a Prefix-Average Constraint). *Fix a position $t \in \{1, \dots, T\}$ and write $z_s := \log \rho_s$ for $s \leq t$. Assume there exist finite constants $\Lambda_+, \Lambda_- > 0$ and an inlier threshold $\tau_0 \geq 0$ such that*

$$-\Lambda_- \leq z_s \leq \Lambda_+, \quad \forall s \leq t.$$

Let $B_s \in \{0, 1\}$ be an abnormal-token indicator satisfying the explicit inlier condition

$$B_s = 0 \implies |z_s| \leq \tau_0, \quad \Lambda_+ \geq \tau_0, \quad \Lambda_- \geq \tau_0.$$

Define its prefix rate by

$$\hat{p}_t := \frac{1}{t} \sum_{s=1}^t B_s,$$

For any threshold $\xi \in [0, 1]$, if $\hat{p}_t \leq \xi$, then the prefix-average log-ratio obeys

$$\frac{1}{t} \sum_{s=1}^t \log \rho_s \in \left[-\tau_0 - \xi(\Lambda_- - \tau_0), \tau_0 + \xi(\Lambda_+ - \tau_0) \right].$$

In particular, choosing log-thresholds (τ_-, τ_+) that satisfy

$$\tau_- \geq \tau_0 + \xi(\Lambda_- - \tau_0), \quad \tau_+ \geq \tau_0 + \xi(\Lambda_+ - \tau_0)$$

ensures $\hat{p}_t \leq \xi$ implies the prefix-average constraint $\frac{1}{t} \sum_{s \leq t} \log \rho_s \in [-\tau_-, \tau_+]$, and hence $\rho_t^{\text{pre}} \in [e^{-\tau_-}, e^{\tau_+}]$.

Proof. Let $N_t := \sum_{s=1}^t B_s$ so that $\hat{p}_t = N_t/t \leq \xi$. For the upper bound, split the sum into inliers and outliers:

$$\sum_{s=1}^t z_s \leq (t - N_t) \tau_0 + N_t \Lambda_+,$$

since inliers satisfy $z_s \leq \tau_0$ and outliers satisfy $z_s \leq \Lambda_+$. Dividing by t and using $N_t/t \leq \xi$ yields

$$\frac{1}{t} \sum_{s=1}^t z_s \leq \tau_0 + \xi(\Lambda_+ - \tau_0).$$

Similarly, using $z_s \geq -\tau_0$ for inliers and $z_s \geq -\Lambda_-$ for outliers gives

$$\frac{1}{t} \sum_{s=1}^t z_s \geq -\tau_0 - \xi(\Lambda_- - \tau_0).$$

The final implication to ρ_t^{pre} follows from $\log \rho_t^{\text{pre}} = \frac{1}{t} \sum_{s \leq t} \log \rho_s$. $\square$

For the sample-based indicator in Lemma D.9, the composition of the inverse certificate with this prefix-rate bound is checked by `k3_abnormal_prefix_average_bound` in `formal/K3Inlier.lean`. This supplies the inlier condition from the actual proxy threshold while retaining the stated bounds $[-\Lambda_-, \Lambda_+]$ on all log-ratios.

E SUPPORTING IDENTITIES AND COMPARISON DETAILS

The results in this section support the main method and reward certificate. They collect the population surrogate connection, standard divergence identities, and comparisons with alternative correction granularities.

E.1 CENTERED SURROGATE EQUIVALENCE AND ERROR DECOMPOSITION

Proof of Proposition 10.8. Write $Q_t(c_t, a) = \mathbb{E}_{\pi_{\text{roll}}}[R \mid c_t, a]$, $V_t(c_t) = \mathbb{E}_{\pi_{\text{roll}}}[R \mid c_t]$, and $\bar{b}(x) = \mathbb{E}[b \mid x]$. Conditional independence of b and policy normalization give

$$\mathbb{E}_{\pi_{\text{roll}}}[(R - b)\rho_t \mid c_t] = \sum_a \pi_\theta(a \mid c_t)(Q_t(c_t, a) - V_t(c_t)) + V_t(c_t) - \bar{b}(x).$$

The last two terms are independent of π_θ . At $\pi_\theta = \pi_{\text{roll}}$ the advantage sum is zero. Multiplying by the predictable mask $m_t(c_t)$, averaging over rollout contexts, and summing over t proves the first equality. The second follows by replacing b by zero and using $\mathbb{E}_{\pi_{\text{roll}}}[R(\rho_t - 1) \mid c_t] = \sum_a \pi_\theta(a \mid c_t)(Q_t(c_t, a) - V_t(c_t))$. On post-EOS padding states the advantage is zero and the ratio is one. $\square$

The finite conditional-law proof, including rollout continuation sums and fixed context/action-mask weights, is checked by `finite_surrogate_equivalence` and `finite_context_surrogate_equivalence` in `formal/SurrogateBridge.lean`; `finite_independent_baseline_average` integrates an explicitly independent baseline sampling law, and `finite_surrogate_derivative_equivalence` transfers the derivative through the constant offset. This gives an exact population connection for the stated finite conditional model; it does not equate individual Monte Carlo estimates. Random-length batch reductions are treated separately in `????`; a general measure-theoretic interface remains outside this formalization.

The finite-state TV step behind Eq. equation 5 is the elementary bound $|\mathbb{E}_{d_t^{\pi_\theta}}[g_t] - \mathbb{E}_{d_t^{\pi_{\text{roll}}}}[g_t]| \leq 2\|g_t\|_\infty \|d_t^{\pi_\theta} - d_t^{\pi_{\text{roll}}}\|_{\text{TV}}$; its finite-sum form is checked by the Lean theorem `finite_tv_expectation_bound`, and summing this step over a finite horizon is checked by `finite_error_decomposition_bound` in `formal/ErrorDecomp.lean`. The finite history-tree modules construct the context-visitation laws, conditional token kernels, and corresponding autoregressive identities explicitly. The abstract finite-sum inequality above is checked independently by `finite_error_decomposition_bound`; a general measure-theoretic conditional-expectation interface remains outside the present formalization.

E.2 CITED SUPPORTING RESULTS FOR IS CORRECTION

This subsection records several standard identities and bounds used in our comparisons of importance sampling (IS) correction mechanisms. All statements are adapted from existing trust-region analyses for LLM-RL (e.g., Qi et al. (2026); Li et al. (2025)). We include them for completeness, with the finite-model Lean proof locations recorded below.

Remark E.1 (TV as an Expectation of the Ratio Residual). Under the mutual-support Assumption 10.3, for any context c_t ,

$$D_{\text{TV}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta}) = \frac{1}{2} \mathbb{E}_{y_t \sim \pi_{\text{roll}}(\cdot|c_t)} [|\rho_t - 1|], \quad \rho_t = \frac{\pi_{\theta}(y_t|c_t)}{\pi_{\text{roll}}(y_t|c_t)}.$$

See, e.g., Qi et al. (2026). For finite token support, the ratio identity is checked by `tv_ratio_identity_of_reverse_support` in `formal/ProMax.lean`; the normalized autoregressive history-tree instantiation is checked by the path-mass results in `formal/TreeSurrogate.lean` and `formal/PrefixConcentration.lean`. Only the general measurable-space version remains outside the present formalization.

Remark E.2 (Coarse-Graining Lower Bounds). Let $\mathcal{V}$ be the vocabulary and let $\mathcal{C} = \{C_1, \dots, C_m\}$ be any partition of $\mathcal{V}$. For a policy π , define the coarse-grained distribution $\pi^{\mathcal{C}}(\cdot|c_t)$ by $\pi^{\mathcal{C}}(C_j|c_t) := \sum_{v \in C_j} \pi(v|c_t)$. Then for any policies π, π' and any context c_t ,

$$D_{\text{TV}}(\pi(\cdot|c_t) \parallel \pi'(\cdot|c_t)) \geq D_{\text{TV}}(\pi^{\mathcal{C}}(\cdot|c_t) \parallel \pi'^{\mathcal{C}}(\cdot|c_t)),$$

and, if $\pi(\cdot|c_t)$ is absolutely continuous with respect to $\pi'(\cdot|c_t)$,

$$D_{\text{KL}}(\pi(\cdot|c_t) \parallel \pi'(\cdot|c_t)) \geq D_{\text{KL}}(\pi^{\mathcal{C}}(\cdot|c_t) \parallel \pi'^{\mathcal{C}}(\cdot|c_t)).$$

See, e.g., Qi et al. (2026). For a finite vocabulary and finite partition map, the TV contraction in this remark is checked by `REINFORCEProMax.finite_coarse_tv_lower_bound` in `formal/CoarseGrain.lean`. The KL contraction and its full equality condition, including zero-mass cells, are checked by `finite_coarse_kl_contraction` and `finite_coarse_kl_equality_iff_proportional` in `formal/CoarseKL.lean`; only the stated one-sided absolute continuity is required.

Remark E.3 (Binary TV Lower Bound from a Sampled Token). Fix a context c_t and a token $y_t \in \mathcal{V}$ such that $\pi_{\text{roll}}(y_t|c_t) > 0$. Consider the binary partition $\mathcal{C} = \{\{y_t\}, \mathcal{V} \setminus \{y_t\}\}$. Then

$$D_{\text{TV}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta}) \geq |\pi_{\text{roll}}(y_t|c_t) - \pi_{\theta}(y_t|c_t)| = \pi_{\text{roll}}(y_t|c_t) |\rho_t - 1|.$$

For a finite token support, this binary-partition instance is checked by `binary_tv_lower_bound` in `formal/CoarseGrain.lean`; the arbitrary finite-partition TV and KL statements are verified in that module and `formal/CoarseKL.lean`, respectively.

Remark E.4 (Sequence-Level TV Bound by Expected Token-Level TV). Fix a prompt x , a finite vocabulary, and normalized autoregressive policies that may depend on the complete prefix context. Write $P^{\pi}(\cdot|x)$ for the resulting distribution over length- T responses. Then

$$D_{\text{TV}}(P^{\pi_{\text{roll}}}(\cdot|x) \parallel P^{\pi_{\theta}}(\cdot|x)) \leq \sum_{t=1}^T \mathbb{E}_{c_t \sim d_t^{\pi_{\text{roll}}}} [D_{\text{TV}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta})].$$

Here d_t^{π} is the distribution of the length- $(t-1)$ prefix context before token t (so the initial context is d_1^{π} and the terminal length- T response distribution is d_{T+1}^{π}). The one-step estimate is

$$\|d_{t+1}^{\pi_{\theta}} - d_{t+1}^{\pi_{\text{roll}}}\|_{\text{TV}} \leq \|d_t^{\pi_{\theta}} - d_t^{\pi_{\text{roll}}}\|_{\text{TV}} + \bar{D}_t.$$

Summing from the common initial context proves the displayed bound. The weighting by roll-out visitation is retained; a uniform maximum over contexts is not required. Both statements are proved directly for the generated path probabilities by `prefixTV_step_weighted` and `sequenceTV_le_expected_tokenTV` in `formal/TreeTV.lean`. No policy-support condition is needed. Shared absorbing EOS padding is allowed. See also Qi et al. (2026).

Remark E.5 (Coarse-Graining Gap for TV). Fix a context c_t and a partition $\mathcal{C} = \{C_1, \dots, C_m\}$. For policies π, π' , define the cell-wise slack

$$s_j(c_t) := \sum_{v \in C_j} |\pi(v|c_t) - \pi'(v|c_t)| - |\pi(C_j|c_t) - \pi'(C_j|c_t)|.$$

Then $s_j(c_t) \geq 0$ and the gap between the true and coarse-grained TV divergences satisfies

$$D_{\text{TV}}(\pi(\cdot|c_t) \parallel \pi'(\cdot|c_t)) - D_{\text{TV}}(\pi^{\mathcal{C}}(\cdot|c_t) \parallel \pi'^{\mathcal{C}}(\cdot|c_t)) = \frac{1}{2} \sum_{j=1}^m s_j(c_t).$$

Moreover, each slack term is bounded by the total mass of its cell:

$$0 \leq s_j(c_t) \leq \pi(C_j|c_t) + \pi'(C_j|c_t).$$

See, e.g., Qi et al. (2026). For finite supports, the displayed slack decomposition, its non-negativity, and the cell-mass bound are checked by `finite_tv_gap_slack_decomposition`, `finite_tv_gap_slack_nonneg`, and `finite_tv_gap_slack_mass_bound` in `formal/CoarseGrain.lean`.

Remark E.6 (Equality Condition for Coarse-Grained KL). Under the absolute continuity condition in Remark E.2, equality in the KL coarse-graining inequality holds if and only if, for every cell C_j with $\pi(C_j|c_t) > 0$, there is a constant $\lambda_j > 0$ such that

$$\pi(v|c_t) = \lambda_j \pi'(v|c_t) \quad \text{for every } v \in C_j.$$

Equivalently, the two conditional distributions inside each positive-mass cell agree after normalization. The condition must include tokens with $\pi(v|c_t) = 0$ but $\pi'(v|c_t) > 0$; restricting the ratio test to the positive support of π is insufficient under one-sided absolute continuity. For arbitrary finite cells, including zero-probability tokens, the full partition characterization is checked by `finite_coarse_kl_equality_iff_proportional` in `formal/CoarseKL.lean`. The proportionality constant is necessarily $\lambda_j = \pi(C_j|c_t)/\pi'(C_j|c_t)$. Cells with $\pi(C_j|c_t) = 0$ contribute zero to both KL expressions and impose no condition on π' . The concrete two-token counterexample `coarse_grain_kl_positive_support_condition_counterexample` in `formal/KLEquality.lean` records why the weaker wording fails.

Definition E.7 (A Finite-Horizon Surrogate for Sequence Rewards). For this finite-horizon statement, fix a deterministic horizon T . If responses terminate at different lengths, we pad them with an absorbing end-of-sequence token and include the corresponding action mask. The post-termination transition is deterministic and shared by both policies (hence its ratio is 1), so the identities below apply to the padded horizon without adding spurious likelihood-ratio terms. Define the rollout-based, sequence-reward surrogate

$$\mathcal{L}_{\pi_{\text{roll}}}^{\text{LLM}}(\pi_{\theta}) := \mathbb{E}_{x \sim P(x), y \sim P^{\pi_{\text{roll}}}(\cdot|x)} \left[R(x, y) \cdot \sum_{t=1}^T (\rho_t - 1) \right],$$

and the remainder term

$$\Delta^{\text{LLM}}(\pi_{\text{roll}}, \pi_{\theta}) := \mathbb{E}_{x \sim P(x), y \sim P^{\pi_{\text{roll}}}(\cdot|x)} \left[R(x, y) \cdot \sum_{t=1}^T (\rho_t - 1) \left(1 - \prod_{j=t+1}^T \rho_j \right) \right],$$

where $\rho_t = \pi_{\theta}(y_t|c_t)/\pi_{\text{roll}}(y_t|c_t)$ and $P^{\pi_{\text{roll}}}(\cdot|x)$ denotes the rollout trajectory distribution. See, e.g., Qi et al. (2026).

Interpretation of the surrogate-remainder split. For a fixed prompt, write $p(y) := P^{\pi_{\text{roll}}}(y|x)$ and $q(y) := P^{\pi_{\theta}}(y|x)$. Under the finite-horizon mutual-support convention, $q(y)/p(y) = \prod_{t=1}^T \rho_t(y)$. Thus the left-hand side of the performance difference identity is $\sum_y q(y)R(x, y) - \sum_y p(y)R(x, y)$, the change in expected sequence reward. The product telescoping identity gives

$$\prod_{t=1}^T \rho_t - 1 = \underbrace{\sum_{t=1}^T (\rho_t - 1) \prod_{j=t+1}^T \rho_j}_{\text{Surrogate}(y)} = \underbrace{\sum_{t=1}^T (\rho_t - 1) \left(1 - \prod_{j=t+1}^T \rho_j \right)}_{\text{Remainder}(y)}.$$

Multiplying by $p(y)R(x, y)$ and summing over trajectories yields

$$\sum_y q(y)R(x, y) - \sum_y p(y)R(x, y) = \sum_y p(y)R(x, y) \text{Surrogate}(y) - \sum_y p(y)R(x, y) \text{Remainder}(y).$$

The surrogate is therefore the additive, rollout-estimable part, while the remainder records the interaction with ratios at later positions. This is an exact finite-horizon algebraic identity, not a claim that the remainder is always nonnegative or negligible. It vanishes for $T = 1$; for longer responses it is small only under additional policy-coupling or trust-region conditions. By Proposition 10.8, this sequence-reward surrogate equals the standard per-step surrogate, and also equals the unclipped trajectory-level surrogate after subtracting its rollout-policy value. Independent baselines do not break this equality. The constructed finite-history-tree version of this identity is checked by `trajectory_surrogate_remainder_identity` in `formal/TreeSurrogate.lean`; the abstract finite-support algebra is checked by `finite_horizon_surrogate_identity`. The additional effects of the actual ProMax gate, normalization, clipping, and batch reduction are identified separately in ??.

Remark E.8 (A Finite-Horizon Performance Difference Bound). Under the mutual-support Assumption 10.3 and the fixed/EOS-padded horizon convention in Definition E.7 (with finite-length Assumption 10.5), one has the exact identity

$$J(\pi_\theta) - J(\pi_{\text{roll}}) = \mathcal{L}_{\pi_{\text{roll}}}^{\text{LLM}}(\pi_\theta) - \Delta^{\text{LLM}}(\pi_{\text{roll}}, \pi_\theta).$$

This display is conditional on the shown expectations being well-defined and on the conditional change-of-measure identity being integrable under the prompt law $P(x)$. Here the Lean-certified statement is for a finite-vocabulary, normalized autoregressive history tree, pointwise in a fixed prompt. It extends to a finite empirical prompt distribution by a finite weighted sum; an arbitrary prompt measure requires the usual integrability and change-of-measure hypotheses in addition to the finite conditional model. The displayed identity is the finite-horizon change-of-measure statement; its telescoping proof uses mutual support so that $\mathbb{E}_{\pi_{\text{roll}}}[R \prod_{s=1}^T \rho_s] = \mathbb{E}_{\pi_\theta}[R]$. For each fixed prompt, a finite vocabulary and a uniform bound $D_{\text{TV}}^{\text{tok}}(c_t) \leq \epsilon$ at every history, the finite autoregressive-tree recursion gives the following quadratic estimate without an additional ratio-coupling premise:

$$|\Delta^{\text{LLM}}(\pi_{\text{roll}}, \pi_\theta)| \leq 2R_{\max} T(T-1) \epsilon^2,$$

This estimate is checked by `surrogate_error_quadratic` in `formal/AutoregressiveTree.lean`; the proof allows history-dependent policies and uses rollout occupancy for the surrogate. The same finite-tree recursion also gives the expected-sum-TV estimate

$$|\Delta^{\text{LLM}}(\pi_{\text{roll}}, \pi_\theta)| \leq 4R_{\max} \sum_{t=1}^T \bar{D}_t.$$

Consequently, whenever the quadratic estimate holds, and also whenever the separate linear estimate holds,

$$J(\pi_\theta) - J(\pi_{\text{roll}}) \geq \mathcal{L}_{\pi_{\text{roll}}}^{\text{LLM}}(\pi_\theta) - \min \left(2R_{\max} T(T-1) \epsilon^2, 4R_{\max} \sum_{t=1}^T \bar{D}_t \right).$$

Both estimates are formalized for the finite autoregressive tree under the stated bounded-reward and policy-normalization conditions, and the finite prompt mixture is handled by linearity. The exact change-of-measure identity remains valid independently of either bound. Theorem 10.10 further supplies the complete position-weighted Adaptive bound on this same remainder. The random-length reduction identities and finite sampling margins are handled separately in ????

E.3 COMPARISON WITH OTHER ADVANTAGE ESTIMATION METHODS

We compare REINFORCE Pro Max with existing critic-free LLM-RL methods along four dimensions: baseline type, advantage normalization strategy, IS correction granularity, and whether the IS correction preserves the causal structure of autoregressive generation. We organize the comparison

into two parts: (i) baseline and normalization choices (this subsection) and (ii) IS correction mechanisms (the next subsection). These dimensions correspond to the key design choices that affect the two factors in the error decomposition (Equation (5)): the advantage factor (targeted by baseline and normalization) and the context shift factor (targeted by IS correction).

Baseline and normalization. REINFORCE++ (Hu et al., 2025) and GRPO (Shao et al., 2024) include the current reward in their group mean baselines. RLOO (Ahmadian et al., 2024) supplies a baseline conditionally independent of each sample’s own response given the prompt, with the exact $(n - 1)/n$ relation to the mean-baseline estimator described in Appendix C.1. Normalization is a separate choice: a common positive scale preserves relative magnitudes, while centering can change signs. ProMax’s asymmetric scales solve the token-weighted mean and second-moment constraints while preserving advantage signs, including when response lengths differ.

E.4 COMPARISON WITH OTHER IMPORTANCE SAMPLING CORRECTION METHODS

We adopt a trust-region masking viewpoint: when mismatch between π_{roll} and π_{θ} is large, one should attenuate or reject unreliable gradient contributions. Existing methods differ mainly in the granularity of this attenuation (see Section 10.6 and Table 1 for a unified notation).

Single-sample ratio statistics vs. distributional drift. Although the per-token ratio residual $|\rho_t - 1|$ is a single-sample quantity, it admits a clean divergence interpretation. Conditional on the context c_t , Remark E.1 implies that $|\rho_t - 1|$ has expectation $2D_{\text{TV}}^{\text{tok}}(c_t; \pi_{\text{roll}}, \pi_{\theta})$ under $y_t \sim \pi_{\text{roll}}(\cdot|c_t)$, so thresholding ρ_t can be viewed as thresholding a noisy proxy for a TV-based trust region. At the same time, the binary lower bound in Remark E.3 shows that the effect of a large ratio depends on the behavior-policy mass at the sampled token: for low-probability tokens, even a large ρ_t can correspond to a small absolute probability shift. Finally, policy drift at the level of full response distributions admits an additive upper bound by the expected token-level TVs along the trajectory (Remark E.4), which motivates per-position and prefix-structured filtering mechanisms.

Finite-horizon remainder bounds. For a finite vocabulary, the autoregressive-tree recursion gives both a quadratic max-TV bound and a linear expected-sum-TV bound for the higher-order term in the rollout-based performance-difference identity (recorded in Remark E.8). These bounds use the rollout occupancy for the surrogate and do not require a ratio bound. The linear expression $4R_{\text{max}} \sum_{t=1}^T \bar{D}_t$ makes the per-position structure explicit and clarifies why long-horizon stability can benefit from controlling divergence along the trajectory rather than only via a single global trust-region statistic. The finite-tree instances of both inequalities are Lean-verified in ‘formal/AutoregressiveTree.lean’; the finite prompt-mixture extension is checked by `family_remainder_quadratic` and `family_remainder_linear` in `formal/PromptMixture.lean`. The EOS/action-mask interface is checked for a context-predictable absorbing continuation by `trajectory_masked_surrogate_eq` in `formal/TreeSurrogate.lean`. The full Adaptive refinement is checked by `trajectory_remainder_adaptive` and `family_remainder_adaptive` in `formal/AdaptiveBound.lean`. It retains the expected local TV at each position and multiplies it by a bound on the remaining continuation drift (Theorem 10.10). Random-length reduction changes the estimator’s population target as quantified in ??.

Coarse-grained divergence proxies as controllable trust-region gates. Full next-token divergences (TV/KL) are expensive to compute over the entire vocabulary, motivating coarse-grained approximations such as Binary or Top- K schemes. Formally, divergences computed on any partition are lower bounds on the true divergences (Remark E.2). Therefore, a gate that masks whenever a coarse-grained proxy exceeds a fixed threshold is *conservative*: it does not reject updates that are provably within the trust region defined by the corresponding full-vocabulary divergence. The remaining question is when such proxies may under-detect drift.

For TV divergence, the coarse-graining gap admits an explicit slack decomposition and a mass-based upper bound (Remark E.5). In particular, for Top- K style partitions that isolate head tokens and aggregate the tail into a single “other” cell, the approximation error is controlled by the tail probability mass. When this aggregated tail mass is small, Top- K TV can serve as a quantitatively controlled proxy for per-position divergence gating, and such per-position proxies can be aggregated along prefixes to yield a prefix-structured gate that respects the causal order of generation. For KL divergence, equality of coarse-grained KL holds exactly when, in every cell with positive p -

mass, the two distributions are proportional throughout that cell (Remark E.6). The finite identity in `formal/CoarseKL.lean` expresses the full-minus-coarse KL as a sum of nonnegative cell-wise Gibbs gaps; consequently, a quantitative near-equality claim should be stated as small total Gibbs gap unless an additional ratio metric is specified. This identifies when KL computed on a reduced support faithfully represents the full-vocabulary KL.

As a concrete causal instantiation, Appendix D.5 includes a prefix abnormal-token rate gate (Variant IV) based on TRM-style per-token divergence proxies. Under mild boundedness conditions on per-token log-ratios, controlling the prefix abnormal-token rate yields a deterministic prefix-average log-ratio bound (Lemma D.10), making the coarse-grained gate a controllable proxy for prefix-level ratio filtering.

Token-level truncation and masking. Under this view, TIS and IcePop are token-level analogues of trust-region masking: they operate directly on the per-token ratio ρ_t . TIS clamps ρ_t to a fixed interval $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$, retaining a biased but bounded weight at every position. IcePop instead masks out tokens whose ratio falls outside the interval, removing extreme sampled coefficients at the cost of possible selection bias and discarded gradient signal. Both apply a local decision at each position and therefore do not account for the downstream context shift caused by an early deviation (??).

The implementation of `seq-mask-tis` and `reinforce_pro` uses a different separation: the sequence or prefix statistic is a detached gate, while the retained mismatch coefficient is the detached, *unclipped* token weight $\exp(\log \pi_{\text{old}} - \log \pi_{\text{roll}})$. Thus these modes should not be described as TIS clipping. The prefix gate controls where the coefficient is applied; it does not by itself bound the magnitude of an accepted coefficient, so the threshold choice and numerical stability remain part of the method’s implementation contract.

Sequence-level aggregation and gating. Sequence-level corrections aggregate ratios over the whole trajectory. GSPO (Zheng et al., 2025) clips the length-normalized sequence likelihood ratio ρ^{seq} (a geometric mean of token ratios) and applies a single coefficient to all tokens, which aligns with sequence-level rewards but loses positional information and can dilute isolated deviations at long horizons (??). TRM (Li et al., 2025) applies a binary sequence gate (e.g., a max-KL trust-region criterion) and rejects entire trajectories, which is theoretically principled but sacrifices per-position granularity.

Prefix-level (causal) filtering. Token- and sequence-level mechanisms leave a gap for long-horizon generation: token-level methods ignore how an early deviation propagates to future contexts (??), while sequence-level gates share a single decision across all positions. Our prefix mask M_t^{pre} (defined in ??) provides a causal, per-position gate constructed from prefix ratio statistics. Compared with TRM’s sequence-level gate $M^{\text{TRM}}(x, y)$, it preserves positional granularity by masking only the token-level loss terms whose prefixes violate the threshold, while still respecting the autoregressive order. The resulting filtering can propagate early-deviation rejection and detect sustained same-side threshold violations (??).

REINFORCE Pro Max. Taken together with the advantage-estimation choices discussed in this appendix, REINFORCE Pro Max targets all four dimensions discussed in this appendix. The RLOO baseline provides statistical independence (Remark C.1); adaptive normalization preserves advantage sign and, in its nondegenerate ideal form, enforces zero empirical mean and unit empirical variance on non-zero tokens, with explicit moment residuals for the implemented safeguards (?? and Proposition C.6, ??); and prefix cumulative IS provides causal, per-position masking with conditional early/late comparisons and sustained-violation guarantees (??). Their combined effect on the actual batch objective is given by ??: empirical advantage reweighting and prefix filtering contribute separate, explicitly measurable terms.